Prime-Gap Dyadic Series: Perturbations, Exact Criteria and Certificates

Proofs, counterexamples and computations for Erdős Problem #251

Will Cook*

September 2026

Abstract

A sparse perturbation theorem shows that specified coefficient congruences and asymptotic block statistics are compatible with both rational and irrational dyadic sums. For any convergent series with nonnegative integer coefficients, one set of upper Banach density zero supports nonnegative corrections attaining an interval of sums. Any prescribed bound tending to infinity is eventually respected; coefficient and partial-sum residues are eventually preserved, with cutoffs independent of the target. Fixed ordinary pair totals protect the congruences; growing digit ranges keep the later weighted choices large enough to fill the interval. Applied to prime gaps with correction bound $\log(n+3)$, the theorem preserves cumulative growth $n \log n$ and gives vanishing total variation error for blocks of length $o(\log \log X)$ sampled on $[X, 2X)$, without ensuring primality. For the actual series, integral shifts of the complete tails characterise rationality. We give sufficient conditions, counterexamples to weaker coefficient conditions, and integer certificates excluding denominators below $2^{40062} > 10^{12059}$. Irrationality remains unresolved.

1 Introduction

The primes $p_0 = 2, p_1 = 3, \ldots$ define the dyadic series

$$\Pi = \sum_{n \geq 0} \frac{p_n}{2^{n+1}} = 2/2 + 3/4 + 5/8 + 7/16 + \cdots .$$

Its irrationality is Erdős Problem #251 [22], recorded in [1, p. 94], [2, p. 62] and [4, p. 103]. Writing $g_n = p_{n+1} - p_n$, summation by parts gives $\Pi = 2 + \sum_n g_n 2^{-(n+1)}$. We use this identity to compare what can be deduced from properties of the coefficients with what would be needed from the actual prime-gap tails.

Theorem 2.1 shows that a substantial collection of coefficient properties does not determine rationality among comparison sequences. Starting with any convergent nonnegative integer dyadic series, it attains an interval of sums using corrections on one sparse set. That set and every eventual coefficient and partial-sum congruence cutoff are fixed before the target. The family therefore contains both rational and irrational

Authorship and AI use. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

sums, not merely a separately chosen example for each preserved property. The corrections can respect any prescribed bound tending to infinity.

At each stage a fixed ordinary correction total protects the congruence calculations at later stages. Redistributing that total between adjacent indices changes its dyadic contribution. Growing digit ranges compensate for the dyadic decay across the increasingly separated stages. A telescoping estimate then ensures that the remaining range spans every gap between adjacent present choices. With a polylogarithmic correction bound, the empirical block laws remain asymptotically equal for lengths $o(\log\log X)$ sampled in $[X, 2X)$. For prime gaps, polynomial nonconcentration and cumulative growth $n\log n$ also survive. The missing requirement is that the cumulative positions be consecutive primes; neither residues nor these statistics supply it.

The actual-prime arguments, beginning in Section 4, are independent of the construction. Summation by parts and a vanishing boundary condition identify the complete tails. Integral shifts then give exact rationality criteria; two consecutive small differences with unequal gaps give a sufficient condition for irrationality. Section 6 certifies individual differences, while Section 7 excludes finite ranges of rational denominators. Neither finite calculation supplies arbitrarily late witnesses. Section 8 tests weaker coefficient conditions by counterexamples; the appendices retain the integer calculations, further criteria and source concordance.

For reference, the first primes and gaps are

i	0	1	2	3	4	5	6	7	8	9
p_i	2	3	5	7	11	13	17	19	23	29
g_i	1	2	2	4	2	4	2	4	6	2

Thus $g_0 = 1$ and all later gaps are even. This parity restriction will be used to characterise two consecutive tail differences of absolute value less than one. The alternative normalisation with denominator 2^i satisfies

$$\sum_{i=0}^{n-1} \frac{p_i}{2^i} = 2\sum_{i=0}^{n-1} \frac{p_i}{2^{i+1}},$$

so it has the same rationality status. The formal factor-of-two identity uses these same finite sums.

Notation. We write $\mathbb{N} = \{0, 1, \ldots\}$, use φ for Euler's totient function, and let $\operatorname{den} q$ be the positive denominator of $q \in \mathbb{Q}$ in lowest terms. A real number is *integral* if it belongs to $\mathbb{Z}$, and $\operatorname{dist}(x, \mathbb{Z})$ is its distance to the nearest integer. A sequence is eventually periodic with period $h \geq 1$ if $a_{n+h} = a_n$ for every sufficiently large n. A set of indices has density zero if its counting function is $o(x)$, and a property holds for almost all indices if its exceptional set has density zero. A property may hold at arbitrarily large indices even when those indices form a set of density zero. An empty sum is zero.

2 Sparse congruence-preserving perturbations

At indices $n, n+1$, the pairs $(0,6), (2,4), (4,2), (6,0)$ all have ordinary sum 6, while their dyadic contributions are $6, 8, 10, 12$ divided by 2^{n+2}. If the cumulative correction before

the pair is odd, we first add 1 at $n-1$. Choosing the pair then leaves the cumulative residue after the triple unchanged. We repeat this operation with nested moduli eventually divisible by every fixed integer.

Finite changes alone preserve rationality, so we use infinitely many triples. Making their separations tend to infinity gives upper Banach density zero, but also makes each weighted contribution smaller. We compensate by increasing the range of choices inside a pair. The estimate (5) shows that all later ranges together cover the gap between adjacent choices at the current stage. The set containing the supports and the interval of sums are chosen before the value to be attained.

Density and block conventions. All intervals of indices contain integers, and log is natural unless a base is displayed. Upper Banach density zero means

$$\lim_{H\to\infty} \sup_{u\in\mathbb{N}} \frac{|S\cap[u,u+H)|}{H} = 0.$$

For integers $X \geq 1$ and $m \geq 1$, let $\mu_{a,X,m}$ be the empirical probability measure obtained by choosing an integer $n \in [X, 2X)$ uniformly and observing $(a_n, \dots, a_{n+m-1})$. The entries are not rescaled, and equal blocks are counted once for each starting index. We use $d_{\mathrm{TV}}(\mu,\nu) = \sup_B |\mu(B) - \nu(B)|$, with the supremum over all sets of blocks. For the same function of the block, bounded in absolute value by B_0, its means under the two measures differ by at most $2B_0 d_{\mathrm{TV}}(\mu,\nu)$. No rescaling of the coefficients is implicit.

Theorem 2.1 (sparse changes preserving congruences)**.** *Let $a_n \in \mathbb{N}$ and $A = \sum_{n\geq 0} a_n 2^{-n-1} < \infty$. For every cutoff K and every $f : \mathbb{N} \to \mathbb{R}$ tending to infinity, there are a set $S \subseteq [K,\infty)$ of upper Banach density zero and a nondegenerate interval $I \subset (A,\infty)$ with the following property. Every $r \in I$ has the form*

$$r = \sum_{n\geq 0} (a_n + e_n) 2^{-n-1},$$

where $e_n \in \mathbb{N}$ is supported on S, $e_n \leq f(n)$ eventually, and, for every integer $q \geq 1$, there is an N_q, chosen independently of r, such that

$$q \mid e_n \quad \text{and} \quad q \mid \sum_{i<n} e_i \qquad (n \geq N_q).$$

For every $\varepsilon > 0$, the construction can instead be chosen with $e_n \leq (\log(n+3))^{\varepsilon}$ eventually and $|S \cap [X,2X)| = O_\varepsilon(X/\log\log X)$. In that case the empirical distributions of original and corrected unnormalised blocks of length $m(X) = o(\log\log X)$, sampled at the same integer starts in $[X,2X)$, have total variation distance tending to zero, uniformly over r. Precisely the coupling error is at most $m|S\cap[X,2X+m)|/X$; with $\|\Phi\|_\infty \leq B_0$ the bounded-test error is at most twice this quantity times B_0, also for tests depending on the starting index.

The hypotheses allow any nonnegative integer sequence of polynomial growth, but not $a_n = 2^n$, whose dyadic series diverges. The bound on the corrections can grow as slowly as $\log\log(n+3)$. A fixed bound is not possible here: eventual boundedness together with eventual divisibility by every fixed modulus would make the corrections

eventually zero; their fixed total would then have to be divisible by every modulus and hence be zero. Nonnegative corrections could not change the sum. The cutoff may depend on the modulus, but the common set S and all cutoffs are fixed before the target. The support of a correction may occupy only part of S. Upper Banach density zero is stronger than ordinary density zero. For example, the union of the integer intervals $[k!, k! + k)$, $k \geq 3$, has ordinary density zero but upper Banach density one. If $k! \leq x < (k+1)!$, at most $k(k+1)/2$ indices have occurred, so their proportion is at most $k(k+1)/(2k!) \to 0$. Nevertheless each interval is filled and their lengths are unbounded. Upper Banach density zero rules out this concentration in long intervals, however far apart those intervals lie.

Proof. Fixed totals. Use $n_j - 1$ to cancel a cumulative residue and $n_j, n_j + 1$ for a pair with fixed ordinary sum. Varying the pair changes its weighted contribution without altering any later residue to be cancelled. Choose an initial index $n_{-1} \geq K$ and centres with $s_j = n_j - n_{j-1} \geq 4$, so the triples do not overlap. Let M_j be positive integers with $M_j \mid M_{j+1}$, and let D_j bound the digit at stage j. The spacing, moduli and digit bounds will be chosen below.

Let C_j record the ordinary correction before the jth triple. Starting from $C_0 = 0$, prescribe

$$c_j = (-C_j) \bmod M_j \quad (0 \leq c_j < M_j), \qquad C_{j+1} = C_j + c_j + M_j D_j.$$

For integers $0 \leq d_j \leq D_j$, set

$$e_{n_j-1} = c_j, \qquad e_{n_j} = M_j d_j, \qquad e_{n_j+1} = M_j(D_j - d_j), \tag{1}$$

and put $e_n = 0$ elsewhere. The two sums of a triple are

$$\sum_{r=n_j-1}^{n_j+1} e_r = c_j + M_j D_j,$$

$$\sum_{r=n_j-1}^{n_j+1} e_r 2^{-r-1} = c_j 2^{-n_j} + M_j D_j 2^{-n_j-2} + d_j M_j 2^{-n_j-2}.$$

The ordinary total contains no d_j, so the recurrence fixes every c_j in advance. In the weighted total only the last term varies: transferring M_j from $n_j + 1$ to n_j adds

$$w_j = M_j 2^{-n_j-2}.$$

After c_j has been inserted, the cumulative correction is divisible by M_j; the two entries of the pair preserve this divisibility. Fix $q \mid M_J$. At index n_J both the coefficient correction and the cumulative correction are divisible by q. At every later triple, q divides C_j and M_j, so it also divides c_j and both pair entries. The intervening corrections are zero. Hence

$$q \mid e_n, \qquad q \mid \sum_{i<n} e_i \qquad (n \geq n_J).$$

The entry c_J need not be divisible by q, which is why it is placed at $n_J - 1$, before the cutoff. The index n_J depends on q and the chosen moduli, but not on the digits or the value to be attained.

Spacing and size. The choices at stage j are $0, w_j, \dots, D_j w_j$. The covering argument will work if the later ranges total at least w_j. Taking $D_i = 2^{s_i} - 1$ compensates for separation by making each later range a scaled dyadic difference:

$$D_i w_i = (2^{n_i - n_{i-1}} - 1) M_i 2^{-n_i - 2} = M_i (2^{-n_{i-1}-2} - 2^{-n_i - 2}).$$

Because $M_i \geq M_j$ for $i > j$, replacing every later modulus by M_j gives the finite estimate

$$\begin{aligned} \sum_{i=j+1}^{J} D_i w_i &\geq M_j \sum_{i=j+1}^{J} (2^{-n_{i-1}-2} - 2^{-n_i - 2}) \\ &= w_j - M_j 2^{-n_J - 2} \qquad (J > j). \end{aligned} \tag{2}$$

No monotonicity of the weights is needed. Each entry of (1) is at most $M_j 2^{s_j}$, so the size cost of separating the stages is exponential. We must fit this product below the allowance at all three indices, while $s_j \to \infty$ and every fixed positive integer eventually divides M_j.

To allow a nonmonotone bound f, put

$$h(n) = \inf_{m \geq n} \min(f(m), m).$$

This nondecreasing lower envelope tends to infinity even when f oscillates. A bound at n_{j-1} therefore controls all three later coordinates; the cap by m ensures $e_n \leq n$ and hence summability for every digit choice. Enlarge $n_{-1} \geq K$ until $h(n_{-1}) \geq 32$, and set

$$\begin{aligned} k_j &= \max\{k \geq 2 : k! 2^{k+2} \leq h(n_{j-1})\}, \\ M_j &= k_j!, \qquad s_j = k_j + 2, \qquad n_j = n_{j-1} + s_j. \end{aligned} \tag{3}$$

The maximum exists: $k = 2$ is admissible because $2!2^4 = 32$, and $k!2^{k+2} \to \infty$. Since $n_j \to \infty$ and h is nondecreasing and divergent, k_j is nondecreasing and tends to infinity. Thus $M_j \mid M_{j+1}$, every fixed q eventually divides $M_j = k_j!$, and $s_j = k_j + 2 \to \infty$. At every coordinate n of the jth triple, this schedule also gives

$$0 \leq e_n \leq M_j 2^{s_j} \leq h(n_{j-1}) \leq \min(f(n), n).$$

Let $S = \bigcup_{j \geq 0} \{n_j - 1, n_j, n_j + 1\}$. For $R \geq 4$, all sufficiently late centres are at least R apart. Every interval of length H contains at most $3(H/R + 2)$ points from those triples; the earlier triples add a fixed number, independent of the interval's position. Divide by H, take the supremum over positions, then let $H \to \infty$. The limiting upper bound is $3/R$; letting $R \to \infty$ proves upper Banach density zero.

Attaining the interval. The support and congruence requirements are now settled independently of the target. Only the digits remain to be chosen. Define

$$\beta = \sum_{j \geq 0} (c_j 2^{-n_j} + D_j w_j), \qquad F_j = \sum_{i \geq j} D_i w_i.$$

Even zero digits contribute $\beta > 0$; F_j is the maximum variable contribution from stage j onwards. The size bound makes both finite, with $F_0 > 0$ and $F_j \to 0$. The total correction is

$$\sum_n e_n 2^{-n-1} = \beta + \sum_j d_j w_j. \tag{4}$$

Taking $J \to \infty$ in (2) now gives

$$F_{j+1} \geq M_j 2^{-n_j-2} = w_j. \tag{5}$$

These bounds do not yet prove that every point of $[0, F_j]$ is attained. A digit d leaves a remainder in $[0, F_{j+1}]$ precisely for current remainders in

$$[dw_j, dw_j + F_{j+1}], \qquad d = 0, \dots, D_j.$$

These intervals meet or overlap by (5), and their union is $[0, F_j]$ because $F_j = D_j w_j + F_{j+1}$.

Given $x \in [0, F_0]$, start with $\rho_0 = x$. Having chosen the first j digits, choose d_j so that

$$\rho_{j+1} = \rho_j - d_j w_j \in [0, F_{j+1}].$$

The covering just proved makes this possible at every stage. Since $0 \leq \rho_j \leq F_j$ and $F_j \to 0$, we obtain

$$x - \sum_{i=0}^{j} d_i w_i = \rho_{j+1} \longrightarrow 0.$$

Thus every $x \in [0, F_0]$ is attained, and we may take $I = (A + \beta, A + \beta + F_0)$. The target $r \in I$ determines only the digits, through $x = r - A - \beta$; it changes neither S, I nor the cutoff $N_q = n_J$ chosen using $q \mid M_J$. This is the finite-choice covering argument of [27, Lemma 4]: the jth choice set has largest gap w_j and diameter $D_j w_j$, its maxima are summable, and the sum of subsequent diameters is at least w_j. Equality is allowed. Fridy's generalised-base lemma [9, p. 194] and the reciprocal-choice argument in [11, Lemma 5.1] are antecedents. The finite-choice argument uses (5); the required comparison of moduli was checked in (2).

The logarithmic scale. For $f(n) = (\log(n+3))^{\varepsilon}$, the size cost 2^{s_j} leads to the iterated logarithm. We choose separations of order $\log\log n_j$ and reserve part of the allowance for growing moduli. Keep the same triples and $D_j = 2^{s_j} - 1$, and set

$$\begin{aligned} s_j &= \left\lfloor \frac{\varepsilon}{2} \log_2 \log(n_{j-1} + 3) \right\rfloor, \\ k_j &= \max\{k \geq 2 : k! \leq (\log(n_{j-1} + 3))^{\varepsilon/4}\}, \end{aligned} \tag{6}$$

with $M_j = k_j!$ and $n_j = n_{j-1} + s_j$. Start far enough out that $s_j \geq 4$ and all choices exist. The exponents $\varepsilon/4$ and $\varepsilon/2$ add to $3\varepsilon/4 < \varepsilon$, so the corrections satisfy the prescribed bound:

$$M_j 2^{s_j} \leq (\log(n_{j-1} + 3))^{3\varepsilon/4} \leq (\log(n+3))^{\varepsilon}$$

at each coordinate n of the triple. After enlarging the initial index, this is also at most n. The moduli remain nested, and every fixed positive integer divides all sufficiently

late moduli. Summability, the congruence induction and (2) therefore apply without change. Since $s_j \asymp_\varepsilon \log\log n_j$, the relevant centres near $[X, 2X)$ are separated by $\gg_\varepsilon \log\log X$. Each contributes at most three support points, giving

$$|S \cap [X, 2X)| = O_\varepsilon(X/\log\log X).$$

Choose the same uniform starting index $n \in [X, 2X)$ for both blocks. They agree when the block avoids S. A changed coordinate $t \in S \cap [X, 2X + m)$ affects only starts in $[t - m + 1, t]$, at most m choices. The union bound gives

$$\begin{aligned} d_{\mathrm{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) &\le \mathbb{P}(e_{n+i} \neq 0 \text{ for some } 0 \le i < m) \\ &\le \frac{m|S \cap [X, 2X + m)|}{X}. \end{aligned}$$

For $m \le X$, two dyadic bands cover $[X, 2X + m)$, giving $O_\varepsilon(m/\log\log X)$. A block of length $o(\log\log X)$ therefore avoids S with probability tending to one. The convergence is uniform over targets because the same S contains every correction support. For a test bounded in absolute value by B_0 that also depends on the starting index, the coupled values agree at unchanged blocks and differ by at most $2B_0$ elsewhere. Their mean difference is consequently at most $2B_0 m|S \cap [X, 2X + m)|/X$. This uses the common starting index, not merely the marginal distributions. No limiting distribution, randomness or independence of the original coefficients is assumed. □

Identical marginal distributions alone would not control an index-dependent test. For example, let $a_n = n \bmod 2$ and $b_n = 1 - a_n$. For every even X, their one-coordinate empirical distributions on $[X, 2X)$ are equal, but the test $\mathbf{1}_{\{x = n \bmod 2\}}$ has mean 1 for (n, a_n) and 0 for (n, b_n). This example is not a sparse perturbation; it isolates why the proof couples blocks at the same index.

The estimate gives a sufficient scale, not an optimal threshold. It tends to zero for fixed block lengths and for $m(X) = \lfloor\sqrt{\log\log X}\rfloor$, but gives no vanishing error for m comparable to $\log\log X$. Its error is also absolute, not relative: probabilities $1/\log X$ and 0 differ by $o(1)$ despite the disappearance of the event. Relative preservation would require an error small compared with the event's probability. Finally, agreement on a finite block does not identify its complete weighted tail; later coefficients still contribute.

Corollary 2.2 (target intervals arbitrarily close to the original sum)**.** *For the sparse theorem for an arbitrary sequence, the target interval can additionally be required to lie in $(A, A + \eta)$ for any prescribed $\eta > 0$.*

Proof. In the construction for a general bound f, every nonzero correction satisfies $e_n \le n$. Choose an integer $K' \ge K$ sufficiently large that $\sum_{n \ge K'} n2^{-(n+1)} = (K' + 1)2^{-K'} < \eta$, and start the construction beyond K'. This still leaves the originally prescribed prefix unchanged. The complete interval of corrections has positive lower endpoint and upper endpoint at most this sum. Here η is chosen before the construction, so the common support and interval may depend on η. Within that interval only the digits depend on the target. This does not make every target above A attainable under one fixed size bound. □

3 Context and mathematical dependencies

Tao's forum comment of 7 October 2025 gives the prime-to-gap reduction and proposes sufficiently uniform quantitative control of about $\log\log n$ consecutive gaps as an input to irrationality [23, comment of 7 October 2025]. Theorem 4.3 records the reduction with its endpoint and convergence proof. An elementary prime bound suffices for it. We use the prime number theorem separately, to deduce $P_n \sim n\log n$ in Corollary 8.6.

Conditional prime-pattern results. Land's draft gives conditional irrationality [13, Conjecture 1, Theorem 2], and Ringer's draft gives conditional normality [26]. For each fixed integer base $b \geq 2$, Ringer's number is $\sum_{n\geq 0} p_n b^{-(n+1)}$ and its normality is to base b. Varying b varies the number, so the result makes no assertion of absolute normality for one constant. The authors supply formalisation material, which we have not rebuilt.

The uniform Hardy–Littlewood hypothesis in [12, Conjecture 1.3] counts prime translates of each nonempty admissible tuple of $k \leq (\log\log x)^3$ distinct integer shifts in $[0, (\log x)^2]$. Admissibility means that the shifts omit a residue class modulo each prime. The main term is the singular series of the tuple multiplied by $\int_2^x (\log t)^{-k}\,dt$, with absolute error at most $Cx^{1-\delta}$. The constants $C, \delta > 0$ and the starting threshold precede the choice of x and of the tuple. Replacing the integral by its leading asymptotic can exceed this error, and qualitative asymptotics for individual fixed tuples do not give the stated uniformity.

In the version rechecked on 18 September 2026, Ringer also states weaker hypotheses for the joint positions of the first L primes after a prime basepoint, with L of order $\log\log X$. The basepoints are uniform among primes in $(X, 2X]$, where X measures prime values. One hypothesis controls a weighted sum of adverse tuple-count errors. The adverse sign alternates with the order of inclusion–exclusion, so both undercounts and overcounts can matter. Another hypothesis allows bounded domination by a comparison law in place of total variation approximation. Neither is a single-gap marginal estimate or a consequence of an upper sieve bound alone. We assume none of these hypotheses here.

Our index band $[X, 2X)$ corresponds to $[p_X, p_{2X})$. The prime number theorem gives $\pi(2p_X) = 2X + o(X)$, so its symmetric difference with the prime basepoints in $(p_X, 2p_X]$ has size $o(X)$. Each set has $X + o(X)$ elements. The two uniform measures have common mass equal to the intersection size divided by the larger cardinality, and hence have total variation distance $o(1)$. This compares the same uniformly bounded observable with the same normalisation at common basepoints. It supplies no relative or quantitative control of the growing patterns required by the preceding conditional results.

Related irrationality problems. Erdős's 1958 article states that $\sum_{n\geq 1} p_{n-1}^k/n!$ is irrational for each $k \geq 1$, and prints a proof only for $k = 1$, explaining that the higher-power proof is more complicated [1, pp. 93–95]. Schlage-Puchta later proved the stronger linear independence over $\mathbb{Q}$ of $1, S_0, S_1, \ldots$, where $S_k = \sum_{n\geq 1} p_{n-1}^k/n!$ [18, Theorem 3]. The fixed dyadic denominator question is stated separately in [4, p. 103], including the difficulty of the case $k = 1$.

The same page proposes irrationality of $\sum_{n\geq 1} p_{n-1}/(d_1 \cdots d_n)$ whenever $d_n \geq 2$ and $d_n = o(p_{n-1})$. Kovač's note of 15 April 2026 disproves this variable-denominator asser-

tion with a sum equal to 1 [24, Theorem 1 and proof, pp. 1–2]. Its printed attribution is ChatGPT 5.4 Pro, orchestrated by Vjeko Kovač. Section 3 of the 1958 article also concerns variable product denominators [1, pp. 96–97]. We make no quantified reinterpretation of its nonstandard asymptotic notation in condition (5), which is unused here. The endpoint example can be read directly: with $G_0 = 1$ and $G_n = \prod_{j=1}^{n}(p_{j-1} + 1)$,

$$\frac{p_{n-1}}{G_n} = \frac{1}{G_{n-1}} - \frac{1}{G_n}, \qquad \sum_{n\geq 1} \frac{p_{n-1}}{G_n} = 1.$$

These examples leave fixed dyadic denominators unchanged as a separate question.

A different dyadic series with bounded coefficients is due to Erdős and Pomerance. If $P(m)$ is the largest prime factor of m, they proved irrationality of

$$\sum_{m\geq 2} \mathbf{1}_{\{P(m)>P(m+1)\}} 2^{-m} \quad [3, §7, p. 320].$$

Erdős and Graham record the complementary indicator [2, p. 62]. Since consecutive integers cannot have the same largest prime factor, its series is $1/2$ minus the displayed one and is also irrational. These binary coefficients differ from the unbounded numerators of Π. Schlage-Puchta's concatenation theorem [18, Theorem 2, p. 1] concerns a base $b \geq 2$ that is not a proper power, with its stated monotonicity and growth assumptions. Concatenation places blocks according to their lengths, whereas our coefficients occupy fixed dyadic positions. We do not use that theorem. The analytic input used below is his Lemma 4, proved by Selberg's sieve: for every fixed nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$, $F(g_n, \dots, g_{n+k}) \neq 0$ for almost all n [18, Lemma 4, pp. 5–6].

Interval covering and divisibility. The bounded perturbation in Theorem 8.1 is a binary expansion. The sparse construction uses the finite-choice lemma of Crmarić and Kovač [27, Lemma 4]. For finite nonnegative choice sets with at least two elements and summable maxima, their lemma fills a single interval if each largest successive gap is at most the sum of the subsequent diameters. Here the sets are $\{0, w_j, \dots, D_j w_j\}$, and the condition is $w_j \leq \sum_{i>j} D_i w_i$. The proof checks this inequality at every stage. Fridy's lemma [9, Lemma, p. 194] assumes nonincreasing weights, a restriction avoided by the finite-choice form. Kovač and Tao use reciprocal choices [11, Lemma 5.1, Theorem 2.5].

Van Doorn and Kovač combine finite subsum covering with a distinguished subsequence of denominators [20, Lemma 7, Proposition 8]. Each distinguished denominator is divisible by all earlier denominators, and every positive integer divides some denominator. Thus every fixed integer divides all sufficiently late distinguished denominators, as it does our factorial moduli. Their condition is imposed on the distinguished subsequence, not on every denominator. At a distinguished cutoff the finite sums lie in the reciprocal grid of the last denominator. Adding the covering inequalities between successive cutoffs gives Lemma 7 for the full finite prefix, leaving no gaps in that grid. Once the denominator of a rational target divides the last denominator and the target lies in the available range, it has a finite representation. Their proposition fills a half-open interval of rational targets by finite subsums. Our construction fills a fixed interval of real targets by infinite choices.

Van Doorn's exchange $\{21d, 28d\} \leftrightarrow \{20d, 30d\}$ preserves the reciprocal sum $1/(12d)$ while changing the ordinary sum from $49d$ to $50d$ [29, proof of Theorem 3]. The pair in (1) preserves the ordinary sum and changes the weighted one. The analogy concerns the conserved quantity, and does not invoke his partition theorem. The present argument adds no new interval-covering principle.

Binary subsums. The *achievement set* of a positive summable sequence consists of sums obtained by retaining or omitting each term. The sequence $(2^{-n})_{n\geq 1}$ fills $[0, 1]$. The sequence $(3^{-n})_{n\geq 1}$ omits $(1/6, 1/3)$ because its terms after the first sum to $1/6$. Nevertheless a failure of this simple overlap test at some indices does not exclude interior elsewhere in the achievement set. Bartoszewicz, Filipczak and Szymonik obtain interior from a central run of integer block sums in a multigeometric family [32, Theorem 2]. Prus-Wiśniowski and Ptak construct Cantorvals with density-zero overlap indices $\{n : a_n \leq \sum_{i>n} a_i\}$ [31, Theorem]. These compact sets are the closures of their interiors, and the endpoints of every nondegenerate interval component are accumulation points of one-point components.

The surrounding classification is discussed in [30, §1] and in Nitecki's exposition [36, Theorem 14], which credits Guthrie and Nymann. The cited preprint's title and theorem numbers differ from those in the published Monthly article. Even prescribing the exact inequality pattern leaves the presence of interior undetermined. Marchwicki and Miska's construction [38, Theorem 2.1], with the proof repair in [37], realises any infinite strict term-dominating index set with a Cantor achievement set. The repair preserves uniqueness, and the latter paper gives a simpler proof of a weaker conclusion without uniqueness. Nowakowski's sufficient condition [39, Definition 2, Theorem 3.1] instead requires every stage of the Star Procedure's positivity recursion to succeed. Knowing that neither term–tail inequality holds eventually therefore cannot replace the overlap calculation in our variable-digit proof.

Uniformity and finite-state comparisons. Adamczewski, Drmota and Müllner compute logarithmic densities of each fixed automatic sequence along primes [28, Theorems 1.2, 1.4]. Their Theorem 1.2 gives a sufficient condition for natural densities to exist and be rational, while Theorem 1.4 reduces logarithmic-density calculation to integers coprime to a fixed modulus. The general logarithmic densities need not be rational. Applying these results to our tails would require a specified finite-alphabet automaton computing the observable, together with uniform error bounds if its state space grows. The theorems themselves concern neither unbounded consecutive gaps nor a growing family of carry automata.

Kuperberg's singular-series estimate permits a specified growing cardinality [12, Theorem 1.1]. Her arithmetic-progression and smooth-weight estimates [33] provide a fixed-parameter comparison, without giving a law for ordered consecutive-gap blocks. Jha's revised preprint [34] studies growing short-interval prime counts under a strong Hardy–Littlewood hypothesis. Such counts alone do not control the weighted consecutive-gap differences in our criteria. A fixed-dimensional limit theorem likewise leaves the required growing-block errors to be proved. At the cited revision, the *Formal*

Conjectures repository leaves Problem #251 unproved [25]. Its zero-based sum starts with the zeroth prime divided by 2^0, and is twice our normalisation.

The tail argument. An ordinary binary expansion is rational exactly when its digits are eventually periodic. Applying carries to integer coefficients changes the sequence to which this criterion refers. Even bounded nonperiodic coefficients can give a rational sum, as Proposition D.6 shows. A classical alternative is the growth criterion for Ahmes series $\sum_k 1/a_k$ with strictly increasing positive integer denominators: $a_k^{1/2^k} \to \infty$ suffices for irrationality [6][11, §1]. Replacing divergence by any sufficiently large fixed threshold fails, since shifted Sylvester sequences grow like C^{2^k} for arbitrarily large C and have rational reciprocal sum. The criterion is sufficient and is not necessary. Splitting $p_i/2^{i+1}$ into p_i equal unit fractions introduces repetitions, and even the distinct denominators 2^n satisfy $(2^n)^{1/2^n} \to 1$.

We instead track reduced tail denominators. Rationality forces an eventually integral shift, not repeating coefficients: the telescope with $K_n = n$ in Proposition 8.10 has coefficients $n-1$ and sum zero. Parity then gives a sufficient test using two small differences. Its occurrence arbitrarily late for every positive shift remains unproved, as does its necessity.

4 Summation by parts, with the endpoint retained

We return to the actual prime gaps $g_n = p_{n+1} - p_n$, with $p_0 = 2$ and $g_0 = 1$. We write

$$\Pi = \sum_{n\geq 0} p_n 2^{-n-1}, \qquad S = \sum_{n\geq 0} g_n 2^{-n-1}.$$

Here S denotes the gap sum. The set of indices used in Section 2 will not be needed below. The short paper calls this same sum G. The two calculations below have different jobs: summation by parts gives $\Pi = 2 + S$, and deleting the first gap gives the initial tail $T_0 = 2S - 1$.

We begin with a finite identity, so no convergence hypothesis is needed. Its endpoint identifies the estimate required before passing to the infinite series. The linked definitions are the dyadic partial sum and the weighted sum of consecutive differences. The formula below writes both sums explicitly.

Proposition 4.1 (finite summation by parts)**.** *For every rational sequence P and every $n \geq 0$,*

$$\sum_{i=0}^{n} \frac{P(i)}{2^{i+1}} = P(0) + \sum_{i=0}^{n-1} \frac{P(i+1) - P(i)}{2^{i+1}} - \frac{P(n)}{2^{n+1}}.$$

Proof. At $n = 0$ both sides equal $P(0)/2$. When n is replaced by $n+1$, the change on the right is

$$\frac{P(n+1) - P(n)}{2^{n+1}} - \frac{P(n+1)}{2^{n+2}} + \frac{P(n)}{2^{n+1}} = \frac{P(n+1)}{2^{n+2}}.$$

This is the next term on the left, which proves the identity by induction. □

Specialising to $P(i) = p_i$, whose first value is $p_0 = 2$, and writing $g_i = p_{i+1} - p_i$ for the zero-based gaps, gives the reformulation.

Theorem 4.2 (prime-gap reformulation)**.** *Let $p_0 = 2, p_1 = 3, \ldots$ be the primes in increasing order and $g_i = p_{i+1} - p_i$. For every $n \geq 0$,*

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

The leading 2 is the first prime, not a normalising constant. At $n = 2$, for instance, the left side is $2/2 + 3/4 + 5/8 = 19/8$ and the right side is $2 + (1/2 + 2/4) - 5/8 = 19/8$.

4.1 The infinite identity and the irrationality equivalence

To pass to the infinite series, let

$$u_n = \frac{p_n}{2^{n+1}}, \qquad v_n = \frac{g_n}{2^{n+1}}.$$

Thus $\sum_n u_n = \Pi$. The termwise identity $v_n = 2u_{n+1} - u_n$ is the dyadic discrete derivative. It expresses each gap term as an integer combination of two consecutive prime terms, so once (u_n) is summable the gap series can be summed after proving absolute convergence, which justifies shifting and subtracting the prime series.

Theorem 4.3 (infinite prime-gap identity)**.** *Both series converge and*

$$\sum_{n\geq 0} \frac{p_n}{2^{n+1}} \;=\; 2 + \sum_{n\geq 0} \frac{g_n}{2^{n+1}}.$$

Proof. The bound $p_n \leq 1250(n+1)^4$ of Appendix A makes (u_n) summable, and $0 < g_n \leq p_{n+1}$ then makes (v_n) summable. The shifted sequence (u_{n+1}) is summable, so summing $v_n = 2u_{n+1} - u_n$ and using $u_0 = 1$ gives

$$\sum_{n\geq 0} v_n = 2\left(\sum_{n\geq 0} u_n - u_0\right) - \sum_{n\geq 0} u_n = \sum_{n\geq 0} u_n - 2. \qquad \square$$

The prime number theorem is needed neither for this convergence nor for the identity. Its asymptotic $p_n \sim n \log n$ is used in Corollary 8.4 and in the argument counting repeated tail values [17, Chapter 6].

Corollary 4.4 (exact irrationality reformulation)**.** Π *is irrational if and only if* $S = \sum_{n\geq 0} g_n 2^{-(n+1)}$ *is irrational. The corresponding zero-based series with denominator* 2^n *equals* $4 + 2S$ *and has the same irrationality status.*

Proof. The identity just proved gives $\Pi = 2 + S$. Adding a rational constant or multiplying by a nonzero rational number preserves rationality and irrationality. $\square$

Concretely, $\Pi = 3.674643966\ldots$ is irrational if and only if $S = 1.674643966\ldots$ is. Neither irrationality statement is established here.

5 The tail recurrence and the exact criteria

We rescale each tail so that its first denominator is again 2. For a convergent series $\sum_{i\geq 0} a_i 2^{-i-1}$ with $a_i \in \mathbb{Z}$, put

$$T_N = 2^{N+1} \sum_{i>N} a_i 2^{-i-1} = \frac{a_{N+1}}{2} + \frac{a_{N+2}}{4} + \cdots.$$

Thus T_N starts with coefficient a_{N+1}, not a_N. Doubling and removing that first coefficient gives $T_{N+1} = 2T_N - a_{N+1}$. Modulo integers, a step is simply doubling; the detailed values of the integer coefficients disappear. This is why the recurrence gives a rationality criterion for arbitrary integer coefficients.

At $N = 0$ we have $T_0 = 2\sum_{i\geq 0} a_i 2^{-i-1} - a_0$. In particular, for the prime gaps $T_0 = 2S - 1 = 2\Pi - 5$, so T_0, S and Π have the same rationality status. We first analyse the recurrence abstractly. We then impose the vanishing boundary condition that singles out complete tails and apply the criterion to these prime-gap tails.

Definition 5.1. Let $g : \mathbb{N} \to \mathbb{Z}$ and $T : \mathbb{N} \to \mathbb{Q}$ or $T : \mathbb{N} \to \mathbb{R}$. Say T satisfies the *dyadic tail recurrence* with *coefficients* g when $T_{N+1} = 2T_N - g_{N+1}$ for every N. Write $\sigma_h(N) = T_{N+h} - T_N$ for the *shift* of length h at N, and call a real number *integral* when it belongs to $\mathbb{Z}$.

For a complete-tail example, take $g_n = 2$ for odd $n \geq 1$ and $g_n = 4$ for even $n \geq 1$. Its complete tail at index 0 is $T_0 = (2/2 + 4/4)/(1 - 1/4) = 8/3$, and the tails alternate between $8/3$ and $10/3$. Thus $\sigma_1(N)$ is always $2/3$ or $-2/3$, whereas $\sigma_2(N) = 0$ for every N. Nonintegrality at one prescribed shift length does not imply irrationality, even for positive even coefficients and genuine tails.

An integral shift returns to the same fractional part. To track the integer accumulated in h steps, define $B_{0,N} = 0$ and $B_{h+1,N} = 2B_{h,N} + g_{N+h+1}$, so that $B_{h,N} = g_{N+1}2^{h-1} + \cdots + g_{N+h}$ is the weighted integer sum accumulated in those h steps.

Theorem 5.2 (block identity)**.** *For every N and h,*

$$T_{N+h} = 2^h T_N - B_{h,N}, \qquad \textit{hence} \qquad \sigma_h(N) = (2^h - 1)\, T_N - B_{h,N}.$$

Proof. The case $h = 0$ follows from $B_{0,N} = 0$. If the first identity holds at h, then

$$T_{N+h+1} = 2(2^h T_N - B_{h,N}) - g_{N+h+1} = 2^{h+1} T_N - B_{h+1,N}.$$

Subtracting T_N gives the second identity. □

Subtracting the tail recurrences at $N + h$ and N gives

$$\sigma_h(N + 1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1}), \tag{7}$$

which is the recurrence for the tail differences. Since $B_{h,N}$ is an integer, Theorem 5.2 also gives the integrality criterion: $\sigma_h(N)$ is integral if and only if $(2^h - 1)T_N$ is integral. For fixed h in this general recurrence, T_N therefore determines whether the shift is integral, regardless of the later integer coefficients. For a complete tail, however, changing those

coefficients also changes T_N; the observation does not permit arbitrary changes to prime gaps.

For rational $T_N = u/v$ in lowest terms, the recurrence gives $T_{N+1} = (2u - g_{N+1}v)/v$. The numerator has greatest common divisor $\gcd(2, v)$ with v, because $\gcd(u, v) = 1$. Thus

$$\operatorname{den}(T_{N+1}) = \frac{\operatorname{den}(T_N)}{\gcd(2, \operatorname{den}(T_N))}, \qquad \sigma_h(N) \in \mathbb{Z} \iff \operatorname{den}(T_N) \mid 2^h - 1. \tag{8}$$

The power of two in the denominator determines how many initial steps must pass before a return is possible; the remaining odd factor d determines the possible return lengths. Once that factor is reached, Euler's congruence supplies $d \mid 2^{\varphi(d)} - 1$. The least positive return length is the multiplicative order of 2 modulo d, taken to be 1 when $d = 1$. Before the power of two is removed there is no return: for example, $T_N = 1/2$ gives no integral shift at N, since $2^h - 1$ is always odd. This periodicity concerns fractional parts, not full tail values or coefficients.

Theorem 5.3 (exact rationality classification)**.** *Let $T : \mathbb{N} \to \mathbb{R}$ satisfy $T_{N+1} = 2T_N - g_{N+1}$ with integer coefficients g. The following are equivalent:*

(i) *T_0 is rational;*

(ii) *$\sigma_h(N)$ is an integer for some $h \geq 1$ and some N;*

(iii) *for some fixed $h \geq 1$, $\sigma_h(N)$ is an integer at every sufficiently large N.*

Consequently T_0 is irrational if and only if every positive-length shift is nonintegral at every index, equivalently if and only if for every $h \geq 1$ and every cutoff some later N has $\sigma_h(N) \notin \mathbb{Z}$.

Proof. For (i)⇒(iii), the block identity makes every T_N rational. By (8), its denominator is a fixed odd integer d for all sufficiently large N. Choose $h = \varphi(d)$; Euler's congruence and the same formula show that every subsequent $\sigma_h(N)$ is integral. The implication (iii)⇒(ii) is immediate. For (ii)⇒(i), the block identity gives $\sigma_h(N) = (2^h - 1)T_N - B_{h,N}$ with $B_{h,N}$ and $\sigma_h(N)$ integers and $2^h - 1 \neq 0$, so T_N is rational, and $T_N = 2^N T_0 - B_{N,0}$ then makes T_0 rational. Negating the three conditions gives the two irrationality formulations. □

Lemma 5.4 (the boundary condition for the tail recurrence)**.** *Let $a_1, a_2, \ldots$ be real numbers with $\sum_{j \geq 1} |a_j| 2^{-j} < \infty$, and let $U_{N+1} = 2U_N - a_{N+1}$. Then*

$$U_N = \sum_{j \geq 1} a_{N+j} 2^{-j} \quad \textit{for every } N \quad \iff \quad 2^{-N} U_N \longrightarrow 0.$$

Proof. Iteration gives $2^{-N} U_N = U_0 - \sum_{j=1}^{N} a_j 2^{-j}$. The limit is zero exactly when U_0 equals the full series; subtracting its first N terms then gives the claimed tail formula. Without the boundary condition, one may add $C 2^N$ to every U_N without changing the recurrence. □

For instance, zero coefficients and initial value $U_0 = 1/12$ give $\frac{1}{12}, \frac{1}{6}, \frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \ldots$, with $U_{N+h} - U_N = (2^h - 1)2^N/12$. Shifts of length 2 become integral at index 2; shifts of length

1 never do. These are not complete tails of the zero sequence, since $2^{-N}U_N = 1/12$ does not tend to zero. The denominator calculation remains valid, but the boundary condition excludes this extra homogeneous term.

The actual prime-gap tails. Put

$$T_N = \sum_{j\geq 1} \frac{g_{N+j}}{2^j},$$

which converges by Theorem 4.3. Reindexing identifies this sum with the rescaled complete tail of S, as in the shifted-gap series identity. Splitting off the first term gives $T_{N+1} = 2T_N - g_{N+1}$ without a rationality hypothesis, as formalised in the real tail recurrence. Since $T_0 = 2S - 1 = 2.349287932\ldots$, the numbers Π, S and T_0 have the same rationality status, and Theorem 5.3 specialises to the actual series: irrationality of Π is equivalent to nonintegrality arbitrarily far out for each positive shift of T. The formal statement is the irrationality equivalence for the actual prime tails. Thus the passage from the series to its complete tails is proved. What remains is to establish the required nonintegrality for these prime-gap tails.

5.1 Pairs of congruent indices and least-common-multiple shifts

Instead of prescribing the offset $M-N$, we may prescribe a positive modulus t and allow any pair with $M \equiv N \pmod{t}$. This gives more freedom in finding a witness, but the test must still work for every t and beyond every cutoff.

Theorem 5.5 (pairs of congruent indices)**.** *S is irrational if and only if for every $t \geq 1$ and every N_0 there are $N, M \geq N_0$ with $M \equiv N \pmod{t}$ and $T_M - T_N \notin \mathbb{Z}$.*

Proof. If S is irrational, take $M = N + t$ for the N supplied by Theorem 5.3 at shift length t. Conversely, suppose S is rational. By (8) the orbit reaches an index N_0 beyond which the reduced denominator is a fixed odd d; let t be the multiplicative order of 2 modulo d, taking $t = 1$ when $d = 1$. For $N, M \geq N_0$ with $M \equiv N$ modulo t, the difference $T_M - T_N$ is then an integer, contradicting the stated property at this t and this cutoff. □

The obstruction under rationality is therefore one particular modulus: the order of 2 modulo the eventual odd denominator. A successful pair for some other modulus does not exclude it.

Equal tail values do not help this criterion, since their difference is zero. Section D.4 shows that rationality and the prime number theorem would in fact force many equal tail values. It also explains why, beyond the rational denominator cutoff, an interval condition excluding integers already contradicts rationality at indices congruent modulo the corresponding multiplicative order. These are ordinary deductions, not additional formalised statements.

Alternatively, one can fix the pairs in advance. Let $L_0 = 1$ and $L_j = \operatorname{lcm}(1, \ldots, j)$. These numbers eventually exceed any starting cutoff and are eventually divisible by any proposed period.

Theorem 5.6 (criterion using least common multiples)**.** *Let $g : \mathbb{N} \to \mathbb{Z}$ and $T : \mathbb{N} \to \mathbb{R}$ satisfy $T_{N+1} = 2T_N - g_{N+1}$. Then T_0 is irrational if and only if $T_{2L_j} - T_{L_j} \notin \mathbb{Z}$ for every $j \geq 0$.*

Proof. Theorem 5.3 gives the forward implication. Conversely, if T_0 is rational then some positive shift length h is integral at every basepoint beyond an index N_0. Choose j so large that $N_0 \le L_j$ and $h \mid L_j$. The telescoping identity $\sigma_{a+b}(N) = \sigma_a(N) + \sigma_b(N+a)$ shows by induction that every positive multiple of h is integral at every such basepoint, so $T_{2L_j} - T_{L_j}$ is an integer. □

One could use factorials instead of least common multiples, since they have the same eventual divisibility property. The least common multiples are no larger at each index. Theorems 5.5 and 5.6 change which tail differences one must test, but remain exact equivalences for integer-coefficient recurrences. Neither proves the needed nonintegrality for the actual prime gaps.

6 A local certificate, and one actual pair

To rule out an integral shift, we can bound two consecutive complete differences in $(-1, 1)$ and check that the corresponding gaps are unequal. Indeed, simultaneous integrality would force both differences to vanish, and their recurrence would then give $g_{N+h+1} = g_{N+1}$. For even gaps we can state these conditions as a single signed interval for the first difference.

Theorem 6.1 (two consecutive small tail differences)**.** *Let T satisfy the dyadic tail recurrence with integer coefficients g, and fix h and N. If*

$$-1 < \sigma_h(N) < 1, \qquad -1 < \sigma_h(N+1) < 1, \qquad g_{N+h+1} \neq g_{N+1},$$

then $\sigma_h(N)$ and $\sigma_h(N+1)$ cannot both be integers. Consequently, if such a pair occurs beyond every threshold, the h-shift is not eventually integral.

Proof. An integer strictly between -1 and 1 is zero. If both shifts were integers, both would vanish, and (7) would give $g_{N+h+1} = g_{N+1}$. The final assertion chooses one such adjacent pair after the alleged onset of integrality. □

Corollary 6.2 (real form and the sufficient condition)**.** *The same statement holds for a real orbit, with the same proof. If for every $h \ge 1$ and every cutoff some later N satisfies the three displayed conditions for the actual prime gaps, then Π is irrational.*

The unequal-gap condition alone is available: at $h = 1$, $N = 2$ it reads $g_4 = 2 \neq 4 = g_3$, and Proposition 6.3 gives such inequalities at arbitrarily large indices for each fixed h. The missing assertion is that both small-tail inequalities and the unequal-gap condition hold at the same arbitrarily late indices. Separate infinite sets of witnesses for the three requirements need not intersect.

Proposition 6.3 (prime gaps do not become periodic)**.** *For every positive h, the actual consecutive-prime-gap sequence is not eventually periodic with period h.*

Proof. For $m \ge 2$ the integers $(m+1)! + 2, \dots, (m+1)! + m + 1$ are composite, so the gaps are unbounded. An eventually periodic sequence of natural numbers has finite range after its preperiod and a bounded initial segment, hence is bounded. □

Far stronger lower bounds for large gaps are known [8, Theorem 1]; the unboundedness of the gaps is all that is needed.

With even gap differences, the three conditions in Theorem 6.1 reduce to two possible signed intervals, each accompanied by a prescribed gap difference. For fixed h write $D_N = \sigma_h(N)$ and $\delta_N = g_{N+h+1} - g_{N+1}$, so that $D_{N+1} = 2D_N - \delta_N$.

Theorem 6.4 (an equivalent signed interval test)**.** *Assume δ_N is even. The conjunction $-1 < D_N < 1$, $-1 < D_{N+1} < 1$, $\delta_N \neq 0$ is equivalent to*

$$(\delta_N = 2 \text{ and } \tfrac{1}{2} < D_N < 1) \quad \text{or} \quad (\delta_N = -2 \text{ and } -1 < D_N < -\tfrac{1}{2}).$$

Proof. Since $\delta_N = 2D_N - D_{N+1}$, the bounds $|D_N| < 1$ and $|D_{N+1}| < 1$ give $-3 < \delta_N < 3$. A nonzero even integer in this interval is 2 or -2. If $\delta_N = 2$, the inequality $-1 < 2D_N - 2 < 1$, together with $|D_N| < 1$, is equivalent to $1/2 < D_N < 1$. Negating both differences gives the case $\delta_N = -2$. Each substitution is reversible. □

For the actual gaps, both indices $N + 1$ and $N + h + 1$ are positive, so δ_N is even for every $N \geq 0$. Theorem 6.4 therefore applies to the actual tails. Its proof uses order, the recurrence and the evenness of δ_N; it does not require rational tail values. This is the real version used in Theorem 8.7.

6.1 An explicit remainder and a certified actual pair

Here N locates the two tails, h separates them, and L is the number of terms retained from each. A finite block supplies the centre of an enclosure, not the full difference. The polynomial prime bound supplies its radius; clearing denominators then makes each test an integer comparison.

Proposition 6.5 (explicit remainder)**.** *For integers $h, N \geq 0$ and $L \geq 1$ put*

$$F_{h,N,L} = \sum_{j=1}^{L} \frac{g_{N+h+j} - g_{N+j}}{2^j}, \qquad P(x) = x^4 + 8x^3 + 36x^2 + 104x + 150,$$

$$E_{h,N,L} = \frac{1250}{2^L}\bigl(P(N + h + L + 2) + P(N + L + 2)\bigr).$$

Then $|\sigma_h(N) - F_{h,N,L}| \leq E_{h,N,L}$. In particular $|F_{h,N,L}| + E_{h,N,L} < 1$ certifies $|\sigma_h(N)| < 1$, and $\operatorname{dist}(F_{h,N,L}, \mathbb{Z}) > E_{h,N,L}$ *certifies $\sigma_h(N) \notin \mathbb{Z}$.*

Proof. Splitting the complete tails after L terms gives

$$\sigma_h(N) = F_{h,N,L} + 2^{-L}\sigma_h(N + L).$$

The last term is the part not determined by the finite gap blocks. The checked bound $p_n \leq 1250(n + 1)^4$ gives $0 \leq g_m \leq p_{m+1} \leq 1250(m + 2)^4$, so its absolute value is at most

$$1250 \sum_{j>L} \frac{(N + h + j + 2)^4 + (N + j + 2)^4}{2^j}.$$

The polynomial identity $2P(x) = (x+1)^4 + P(x+1)$ telescopes to $\sum_{k=1}^{J}(m+k)^4 2^{-k} = P(m) - P(m+J)2^{-J}$, whose last term tends to zero. Apply it with $m = N+h+L+2$ and $m = N+L+2$ after writing $j = L+k$. The two tests follow from the triangle inequality and the definition of distance to $\mathbb{Z}$. □

Proposition 6.6 (a certified adjacent pair)**.** *For the actual prime gaps, $h = 1$ and $N = 2$ satisfy the three hypotheses of Theorem 6.1: both $\sigma_1(2)$ and $\sigma_1(3)$ lie in $(-1,1)$ and are nonintegral, and $g_4 = 2 \neq 4 = g_3$.*

Proof. Take $L = 40$ and $Q = 2^{40} = 1099511627776$. Exact integer arithmetic over the first 46 primes gives

N	$QF_{1,N,40}$	$QE_{1,N,40}$
2	−662838684750	11764181250
3	873345886050	12805761250

In each row $|QF| + QE < Q$, which puts the whole certified interval inside $(-1,1)$, and QE is smaller than the distance from QF to the nearest multiple of Q, which excludes every integer. The margins are 424908761776 and 213359980476 respectively. These are strict integer comparisons rather than inferences from rounded numerical tails. Appendix B replays them. □

This certifies one actual adjacent pair, not a recurrence theorem for such pairs. Irrationality still requires a witness for every positive h beyond every cutoff; the successful indices may depend on h.

Proposition 6.7 (certifying two small differences from one enclosure)**.** *Let $D_{N+1} = 2D_N - \delta_N$ with real D_N, and suppose $\delta_N = 2s$ for $s \in \{-1,1\}$. If integers A, B, Q satisfy $Q > 0$, $B \geq 0$, $|QD_N - A| \leq B$, and*

$$2sA - Q > 2B, \qquad Q - sA > B,$$

then $1/2 < sD_N < 1$, $|D_{N+1}| < 1$, and both D_N, D_{N+1} are nonintegral. In particular, the two small-difference conditions are certified from just one tail enclosure.

Proof. Since $|s| = 1$, the enclosure gives

$$\frac{sA - B}{Q} \leq sD_N \leq \frac{sA + B}{Q}.$$

The first strict inequality says that the lower endpoint exceeds $1/2$; the second says that the upper endpoint is less than 1. Thus $sD_N \in (1/2, 1)$. The recurrence and $\delta_N = 2s$ now give $sD_{N+1} = 2sD_N - 2 \in (-1, 0)$. Both intervals exclude integers, and the second also gives $|D_{N+1}| < 1$. □

For $h = 1, N = 2$, the exact data already used above give $Q = 2^{40}$, $A = -662838684750$, $B = 11764181250$ and $s = -1$. The two positive integer margins are respectively 202637379224 and 424908761776. Proposition 6.6 encloses each difference separately. Here the first enclosure and the exact gap difference give the second interval through the recurrence. The second enclosure is therefore an independent cross-check, not a necessary input.

7 Two lower bounds on a possible rational denominator

Here the enclosure is for Π itself, rather than a tail difference. A finite prime sum gives the lower endpoint; the polynomial prime bound gives the upper endpoint. Two fractions of determinant one bracket this entire interval, so every rational in it has denominator at least the sum of their denominators. This last step is an integer identity, independent of how the fractions were found. Appendix B prints the smaller certificate; the linked Lean development checks the integer comparisons for both. The bounds exclude finite denominator ranges, not rationality.

Theorem 7.1 (a denominator bound checked by exact integer comparisons)**.** *Let $a \in \mathbb{Z}$ and let b be a positive integer. If $\Pi = a/b$ then $b \geq 2^{589} > 10^{177}$, and the same floor holds for every rational equal to S.*

Proof. Put $c = 1229$, the number of primes below 10^4, and $A = \sum_{i=0}^{c-1} p_i 2^{\,c-i-1}$, so that $A/2^c$ is the cth partial sum of Π. All terms are positive, so $A/2^c \leq \Pi$. For the upper bound, $p_{c+j} \leq 1250(c+j+1)^4$. Since $c \geq 9$, each successive ratio of $(c+1+j)^4$ is at most $(11/10)^4 < 3/2$. Hence $(c+1+j)^4 \leq (c+1)^4(3/2)^j$ for all $j \geq 0$. The omitted dyadic terms are therefore bounded by a geometric series with ratio $3/4$, whose sum gives

$$\frac{A}{2^c} \;\leq\; \Pi \;\leq\; \frac{2A + 5000(c+1)^4}{2^{\,c+1}}.$$

The four positive integers u, v, u', v' printed in Appendix B satisfy

$$u'v - uv' = 1, \qquad \frac{u}{v} < \frac{A}{2^c}, \qquad \frac{2A + 5000(c+1)^4}{2^{\,c+1}} < \frac{u'}{v'}, \qquad v + v' \geq 2^{589},$$

all four being integer comparisons after clearing the positive denominators. If $\Pi = a/b$, the strict lower and upper bounds make $av - bu$ and $bu' - av'$ positive integers. Each is therefore at least 1. The determinant identity expresses b as their positive integer combination of v' and v:

$$b = b(u'v - uv') = (av - bu)v' + (bu' - av')v \geq v + v' \geq 2^{589}.$$

The only properties needed of the bracketing fractions are their order, positive denominators and determinant one. Their decimal expansions and the way they were found play no part in this denominator bound. Since $589 \log_{10} 2 > 177$, the floor exceeds 10^{177}. If $S = a/b$, apply the same argument to $\Pi = (a + 2b)/b$. □

The Lean kernel evaluates the four inequalities itself, on the same integers, after recomputing the primes below 10^4 by trial division; no compiled code is trusted. Its analytic input is the same enclosure, derived from the polynomial bound $p_n \leq 1250(n+1)^4$. The four integers u, v, u', v' have 177 and 178 digits.

Theorem 7.2 (certified continued-fraction exclusion)**.** *Every rational equal to Π, and hence every rational equal to S, has reduced denominator $q \geq 2^{39997}$, and therefore $q > 10^{12040}$.*

The same determinant argument gives the stronger floor 2^{40062} below. The smaller constants in the statement and the different continued-fraction pairs are accounted for after the proof.

Proof. Put $c = 80200$, the number of primes below 1 023 068, and $A = \sum_{i=0}^{c-1} p_i 2^{c-i-1}$. The estimate in the proof of Theorem 7.1 holds for every $c \geq 9$ and gives

$$\frac{A}{2^c} \leq \Pi \leq \frac{2A + 5000(c+1)^4}{2^{c+1}}.$$

Four positive integers u, v, u', v', described after this proof, satisfy

$$u'v - uv' = 1, \qquad \frac{u}{v} < \frac{A}{2^c}, \qquad \frac{2A + 5000(c+1)^4}{2^{c+1}} < \frac{u'}{v'}, \qquad v + v' \geq 2^{40062},$$

all four being integer comparisons after clearing the positive denominators. If $\Pi = a/b$ with $a \in \mathbb{Z}$ and b a positive integer, the determinant calculation in the proof of Theorem 7.1 gives

$$b \geq v + v' \geq 2^{40062} > 10^{12059}.$$

The last inequality holds because $10^{12059} = (10^{59})^{204}\, 10^{23}$ and $2^{40062} = (2^{196})^{204}\, 2^{78}$, with $10^{59} < 2^{196}$ and $10^{23} < 2^{78}$. The reduced denominator q is one such b, so $q \geq 2^{40062} > 2^{39997}$ and $q > 10^{12059} > 10^{12040}$. If $S = a/b$, apply the same argument to $\Pi = (a + 2b)/b$. □

The integers u, v, u', v' come from the continued fractions of the two endpoints $A/2^c$ and $(2A + 5000(c+1)^4)/2^{c+1}$, which agree in their first 23407 partial quotients. The fraction u/v is the convergent formed by those partial quotients, and u'/v' is the mediant of u/v and the convergent before it. Only the four comparisons enter the proof. The numerators u, u' have 12061 decimal digits and the denominators v, v' have 12060.

The Lean kernel checks every step of this proof and evaluates every computation itself; no compiled code is trusted. It recomputes c and A from the definition of primality, scanning the integers below 1 023 068 in 250 consecutive blocks and checking the prime count and partial numerator at the end of each block. It decides primality by trial division below 1012. On $[1012, 1013^2)$, which contains the rest of the scan because $1013^2 = 1\,026\,169$, it uses the fact that such an integer is prime exactly when it is coprime to the product of the primes below 1012, and it recomputes that product by trial division. Lean proves that the two primality tests agree at every natural number. The kernel then decides the four comparisons, together with $10^{59} < 2^{196}$ and $10^{23} < 2^{78}$; the enclosure and the determinant step are proved in general form. Lean proves the bound $b \geq 2^{40062}$ and $b > 10^{12059}$ for both series and derives the theorem as stated from it, using only Lean's three standard axioms.

An independent program outside Lean recomputes A by a prime sieve and uses exact integer arithmetic. Its recorded output has the same 23407 common partial quotients. It brackets the interval with the last two consecutive convergents, whose denominator sum has bit length 40062, and checks the advertised bound $2^{39997} > 10^{12040}$. This pair

differs from the convergent–mediant pair in the Lean certificate above, whose denominator sum has bit length 40063 and is at least 2^{40062}. The two bit lengths refer to different valid pairs.

The constants 2^{39997} and 10^{12040} in the statement come from a separate computation with a bracket of width 2^{-80000}, which used the best-approximation property of convergents [10, §6, Theorems 16 and 17]; Short gives an accessible statement and proof [19, Theorem 1.1]. Its recorded output lists the generating program and its digest and a last convergent whose denominator has bit length 39998, hence is at least 2^{39997}; the program also checked its arithmetic against the known expansions of e and π. That bit length describes one convergent only; the bound for every rational in the enclosure comes from the certified sum $v + v'$ above, and $2^{39997} > 10^{12040}$ is a separate integer comparison.

These computations exclude finite denominator ranges. Irrationality would require unbounded exclusions, not just a larger finite bound. Even extending the prefix need not improve the bound at every step. No irrationality-exponent estimate is asserted.

8 Coefficient conditions and counterexamples

The examples test different proposed substitutes for information about the actual tails. Bounded changes preserve a fixed modulus and transfer polynomial nonconcentration through finitely many translates. Sparse changes preserve the larger collection of properties in the main theorem because most sampled blocks are untouched. Later examples prescribe integer tails first and derive coefficients with recurring values or growth conditions. These mechanisms have different conclusions; none constructs consecutive primes.

We also show that the actual small-pair witnesses, if they occur arbitrarily late, must have density zero. This does not rule them out: infinitely many witnesses suffice, and positive density is not required.

8.1 Bounded residue-preserving perturbations

For one fixed positive modulus M, we may add either 0 or M after a prescribed prefix. The weighted choices form a binary expansion, so we can choose a rational sum while preserving every coefficient modulo M. This construction permits changes at every index after the prefix; it imposes neither density-zero support nor eventual preservation of all moduli.

Theorem 8.1 (bounded perturbations preserving congruences)**.** *Let a_n be natural numbers with $\sum_{n\ge 0} a_n 2^{-(n+1)}$ convergent. For every integer $M \ge 1$ and every cutoff K there are digits $\varepsilon_n \in \{0,1\}$, zero for $n < K$, such that $\sum_{n\ge 0}(a_n + M\varepsilon_n)2^{-(n+1)}$ is rational.*

Proof. Write $A = \sum_{n\ge 0} a_n 2^{-(n+1)}$ and choose a rational $r \in (A, A + M2^{-K})$. A binary expansion of $(r - A)/M$ has the form $\sum_{n\ge K} \varepsilon_n 2^{-(n+1)}$ with $\varepsilon_n \in \{0,1\}$; set $\varepsilon_n = 0$ for $n < K$. The perturbed series converges and equals r. □

Hence a property shared by every allowed perturbation cannot by itself force irrationality, since one such perturbation has a rational sum; this consequence is not stated separately in Lean. Lean also checks the construction at the actual consecutive prime gaps,

for every $M \geq 1$ and every K: the perturbed prime-gap series is rational, and each perturbed gap lies in $[g_n, g_n+M]$ and is congruent to g_n modulo M. For each fixed $k \geq 1$, the sum of k consecutive gaps increases by at most kM. Thus infinitely many bounded gaps and bounded clusters of each fixed size survive, but their original numerical bounds need not. Nonnegative corrections also preserve lower bounds for large gaps. Taking $M = 2q$ preserves parity and the cumulative positions modulo a prescribed q, hence any limiting residue frequencies of those positions. These conclusions concern the size and residue information in the results discussed in Section 10; they establish neither primality of the new positions nor preservation of every quantitative assertion of those theorems.

8.2 Algebraic nonconcentration survives the rationalising perturbation

Schlage-Puchta's lemma states that $F(g_n, \dots, g_{n+k}) \neq 0$ for almost all n, for every fixed nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$ [18, Lemma 4, pp. 5–6]. We say that an integer sequence has *fixed-block nonconcentration* if it satisfies this conclusion for every $k \geq 0$ and every such F. A bounded integer perturbation preserves the property: there are only finitely many translated polynomials to consider for each fixed block. No randomness or independence assumption is required.

This nonconcentration condition excludes every bounded integer sequence: if its values lie in a finite set E, the nonzero polynomial $\prod_{c\in E}(x_0 - c)$ vanishes at every index. It also excludes a sequence satisfying a fixed nonzero polynomial relation on every short block, such as $a_n = n$, for which $x_1 - x_0 - 1$ vanishes. The density-zero exceptional set depends on the fixed polynomial. There cannot be one such set for all polynomials: $x_0 - a_n$ vanishes on the block beginning at n, so a common exceptional set would contain every index. No uniform estimate for growing families of polynomials or block lengths is asserted.

In the next proposition, N, H are nonnegative integers, h is an integer, and all counts are over integer indices and integer triples.

Proposition 8.2 (finite counting for shifted gap differences)**.** *Write p_n for the primes indexed from $p_0 = 2$ and $g_n = p_{n+1} - p_n$. For $h \geq 2$ and $r \in \mathbb{Z}$, let $M_{h,r}(N)$ count $n < N$ with $g_{n+h} - g_n = r$. Let $Q_{N,H,r}$ count triples (x, d, s) with $x < p_N$, $0 < d < s \leq H$, $d + r > 0$, and all four integers $x, x + d, x + s, x + s + d + r$ prime. Then, for every $N, H \geq 0$,*

$$(H + 1)M_{h,r}(N) \leq (h + 1)p_{N+h+1} + (H + 1)Q_{N,H,r}.$$

Proof. For a counted index n, put $x = p_n$, $d = g_n$ and $s = p_{n+h} - p_n$. If the total span $p_{n+h+1} - p_n$ is at most H, then $0 < d < s \leq H$, $d + r = g_{n+h} > 0$, and the four primes are precisely $x, x + d, x + s, x + s + d + r$. The count $Q_{N,H,r}$ need not require these primes to be consecutive: it is an upper bound, and $x = p_n$ determines n, making the map injective. For the remaining indices the integer span is at least $H + 1$. Each gap appears in at most $h + 1$ of the spans, so their total is at most $(h + 1)\sum_{i<N+h} g_i < (h + 1)p_{N+h+1}$. Multiply the resulting bound on the exceptional count by $H + 1$ and add the small-span contribution. □

Increasing H reduces the bound for long spans but enlarges the prime-pattern count. To deduce zero density from this finite inequality, for each $\varepsilon > 0$ and all large N one

needs a single choice of H making the entire right side less than $\varepsilon N(H+1)$. For $h = 1$, the same definitions give $s = d$, so $x + d = x + s$. The strict condition $d < s$ in the four-prime count fails; that case requires a three-prime count instead. Zero density itself is already known for every fixed $h \geq 1$ and $r \in \mathbb{Z}$: apply Schlage-Puchta's lemma above to the nonzero polynomial $x_h - x_0 - r$. Thus the finite reduction should not be read as leaving that upper bound open. Neither upper bound supplies the late occurrences or weighted-tail separation required for irrationality.

For the perturbation claim, first suppose that a has fixed-block nonconcentration and every correction is either 0 or a fixed integer $M \geq 1$. For a fixed $r \in \mathbb{Z}$, an equality $b_{n+1} - b_n = r$ then requires $a_{n+1} - a_n \in \{r - M, r, r + M\}$. Each of these three equalities holds on a set of density zero. The general argument uses the same finite-union step for translated polynomials.

Theorem 8.3 (nonconcentration is perturbation-stable)**.** *Let $a : \mathbb{N} \to \mathbb{Z}$ have fixed-block nonconcentration, let $E \subset \mathbb{Z}$ be finite, and let $b_n = a_n + e_n$ with $e_n \in E$ for every n. Then b has fixed-block nonconcentration.*

Proof. Fix $k \geq 0$ and a nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$. For a tuple $\mathbf{e} = (e^{(0)}, \dots, e^{(k)}) \in E^{k+1}$ put $F_{\mathbf{e}}(x_0, \dots, x_k) = F(x_0 + e^{(0)}, \dots, x_k + e^{(k)})$. The substitution $x_i \mapsto x_i + e^{(i)}$ is a ring automorphism of $\mathbb{Z}[x_0, \dots, x_k]$ with inverse $x_i \mapsto x_i - e^{(i)}$, so $F_{\mathbf{e}} \neq 0$. If $F(b_n, \dots, b_{n+k}) = 0$ then $F_{\mathbf{e}}(a_n, \dots, a_{n+k}) = 0$ for the tuple $\mathbf{e} = (e_n, \dots, e_{n+k})$, whence

$$\{\, n : F(b_n, \dots, b_{n+k}) = 0 \,\} \subseteq \bigcup_{\mathbf{e} \in E^{k+1}} \{\, n : F_{\mathbf{e}}(a_n, \dots, a_{n+k}) = 0 \,\}.$$

Each set on the right has density zero by hypothesis, and there are only finitely many such sets because E^{k+1} is finite. Their union therefore has density zero. □

Finiteness of the correction set is essential to this argument; no independence between the corrections and the sequence is needed. Without a restriction on the corrections, the choice $e_n = -a_n$ would give $b_n = 0$, which fails nonconcentration even for the polynomial x_0.

Corollary 8.4 (nonconcentration does not force irrationality)**.** *Fix $M \geq 1$ and $K \geq 0$, and let b be the perturbed sequence supplied by Theorem 8.1 at the actual prime gaps. Then $\sum_{n \geq 0} b_n 2^{-(n+1)}$ is rational, $b_n = g_n$ for $n < K$, $b_n - g_n \in \{0, M\}$ and $b_n \equiv g_n \pmod{M}$ for every n, b has fixed-block nonconcentration, and the cumulative sequence $P_n = 2 + \sum_{i<n} b_i$ satisfies $p_n \leq P_n \leq p_n + Mn$ and hence $P_n \sim n \log n$.*

The Lean proof assumes Schlage-Puchta's Lemma 4 [18] and the prime number theorem.

Proof. Theorem 8.1 gives the rational sum, prefix and congruence assertions. Summing $0 \leq b_i - g_i \leq M$ and using $\sum_{i<n} g_i = p_n - 2$ gives $p_n \leq P_n \leq p_n + Mn$. Now $p_n \sim n \log n$ [17, Chapter 6] implies $P_n \sim n \log n$. The perturbation takes values in the fixed two-element set $E = \{0, M\}$, so Theorem 8.3 applies to the actual gaps, which have fixed-block nonconcentration by Schlage-Puchta's lemma [18, Lemma 4]. □

All the corollary's properties hold for one rational-sum sequence; the positions P_n need not be prime. Taking M even also preserves eventual evenness. In particular $M =$

$2q$ preserves parity and modulus q, with $b_n \le g_n + 2q$. Nonconcentration remains a fixed-polynomial, fixed-block assertion, not a uniform estimate for growing families.

Proposition 8.5 (nonconcentration under sparse changes)**.** *Let $a, b : \mathbb{N} \to \mathbb{Z}$ agree off a set S of ordinary density zero. If a has fixed-block nonconcentration, then so does b. No boundedness assumption on $a - b$ is needed.*

Proof. Fix k and a nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$. The two blocks agree unless $n + i \in S$ for some $0 \le i \le k$. Consequently

$$\begin{aligned}&|\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| \\ &\quad \le |\{n < N : F(a_n, \dots, a_{n+k}) = 0\}| + \sum_{i=0}^{k} |S \cap [i, N + i)|.\end{aligned}$$

The first count is $o(N)$ by nonconcentration for a; the remaining finitely many counts are $o(N)$ because S has density zero. The count concerns the event $F = 0$, not the sizes of F or the corrections. The sequence b is already fixed, so this holds for every fixed F, k, without a uniform estimate for growing polynomial families or block lengths. □

Corollary 8.6 (a rational sum with the stated prime-gap statistics)**.** *For every prescribed finite prime-gap prefix and $0 < \varepsilon \le 1$, there is an altered sequence $b = g + e$, with $P_n = 2 + \sum_{i<n} b_i$, for which one can simultaneously impose a rational dyadic value, nonnegative integer corrections eventually at most $(\log(n+3))^{\varepsilon}$, all fixed eventual coefficient and cumulative congruences, fixed-block polynomial nonconcentration, and vanishing total variation distance between unnormalised block distributions for lengths $o(\log\log X)$. The cumulative positions satisfy*

$$0 \le P_n - p_n = O_\varepsilon\left(\frac{n(\log(n+3))^{\varepsilon}}{\log\log n}\right), \qquad P_n \sim n\log n.$$

Lean proves this for every $\varepsilon > 0$, assuming Schlage-Puchta's Lemma 4 [18] only for the nonconcentration clause and the prime number theorem only for $P_n \sim n\log n$.

Proof. Apply Theorem 2.1 to the actual gaps, choose a rational target in its interval, and fix the resulting $b = g + e$. Proposition 8.5 then transfers Schlage-Puchta's nonconcentration [18, Lemma 4] to this one sequence, for every fixed polynomial and block length. To obtain the cumulative support bound, split $[\sqrt{n}, n)$ into dyadic intervals, use $\log\log x \asymp \log\log n$ there, and bound the first $\sqrt{n}$ indices trivially. Thus $|S \cap [0, n)| = O(n/\log\log n)$. The pointwise correction bound gives the displayed estimate, including a fixed constant for the exceptional prefix. Since $(\log n)^{\varepsilon-1}/\log\log n \to 0$ for $0 < \varepsilon \le 1$, the prime number theorem [17, Chapter 6] gives the last conclusion. The position series is rational as well: reversing the order of summation of nonnegative terms gives

$$\sum_{n\ge 0} \frac{P_n}{2^{n+1}} = 2 + \sum_{i\ge 0} b_i \sum_{n>i} 2^{-n-1} = 2 + \sum_{i\ge 0} \frac{b_i}{2^{i+1}} \in \mathbb{Q}.$$

The finite value on the right also proves convergence on the left. □

The case $\varepsilon > 1$ needs no new construction: use exponent 1. Its corrections satisfy the larger allowance, while the stronger error $O(n \log(n+3)/\log\log n) = o(n \log n)$ still preserves cumulative growth. Thus every positive exponent is available. The construction supplies the support and correction estimates; Schlage-Puchta's lemma supplies nonconcentration of the original gaps, and the prime number theorem supplies their cumulative asymptotic.

For each fixed integer $y \geq 2$, take modulus $y!$. Once the congruence holds and $p_n > y$, $\gcd(P_n, y!) = 1$ excludes every prime divisor at most y. This is stronger than eventual oddness. The cutoff depends on y; primality would require this exclusion up to $\sqrt{P_n}$ at the same index. Even prime-valued positions could omit intervening primes.

The bounded construction transfers nonconcentration by finitely many polynomial translates, not by preserving block frequencies. The sparse construction transfers block laws by leaving most blocks unchanged, with absolute error tending to zero for $m = o(\log\log X)$. Neither argument preserves relative rare-event frequencies or the quantitative prime-pattern hypotheses in Section 3.

8.3 Density of indices with two small tail differences

When the two tail differences have absolute value less than one and the gaps are unequal, the calculation above restricts the gap difference to 2 or -2. For each fixed shift, Schlage-Puchta's lemma gives density zero for both events. We use this to bound the set of possible witnesses, without excluding witnesses at arbitrarily large indices.

Theorem 8.7 (density zero for the small-difference condition)**.** *Fix $h \geq 1$. The set of $N \geq 1$ at which the three hypotheses of Theorem 6.1 hold for the actual prime gaps has density zero. For the same h, the set of N with $g_{N+h+1} = g_{N+1}$ also has density zero.*

The Lean proof assumes Schlage-Puchta's Lemma 4 [18].

Proof. For $N \geq 1$ every gap involved is even, so $\delta_N = g_{N+h+1} - g_{N+1}$ is even. If $|\sigma_h(N)| < 1$, $|\sigma_h(N+1)| < 1$ and $\delta_N \neq 0$, then $\delta_N = 2\sigma_h(N) - \sigma_h(N+1)$ lies in $(-3, 3)$, so $\delta_N = \pm 2$. Apply Schlage-Puchta's lemma [18, Lemma 4] at index $n = N+1$ with $k = h$ to the two polynomials $x_h - x_0 - 2$ and $x_h - x_0 + 2$, neither of which vanishes identically: each of $\{N : \delta_N = 2\}$ and $\{N : \delta_N = -2\}$ has density zero, and the event is contained in their union. The second assertion is the same lemma applied to $x_h - x_0$. □

Unequal gaps occur at almost every index, a stronger conclusion than Proposition 6.3. But combining unequal gaps with the two small tail differences forces their difference to be exactly 2 or -2, which occurs only on a density-zero set. Thus a proof of Problem 10.2 must find arbitrarily large successful indices in that sparse set. A positive-density lower bound for the same event is impossible. An averaging argument that detects a sparse set is not ruled out.

The proof of Schlage-Puchta's lemma gives a quantitative upper bound as well. For fixed h, it bounds the number of eligible indices in $[X, 2X)$ by $O_h(X/\log\log X)$. To see the two contributions, discard starts whose block of $h+1$ gaps contains a gap exceeding $\log X \log\log X$. The sum of these block sums is $O_h(X \log X)$ by the prime number theorem, so there are at most $O_h(X/\log\log X)$ discarded starts. For the remaining starts, the

sieve calculation in the cited proof bounds the zeros of each of $x_h - x_0 - 2$ and $x_h - x_0 + 2$ by

$$O_h\left(\frac{X(\log\log X)^{2h+2}}{\log X}\right) = o_h\left(\frac{X}{\log\log X}\right).$$

The shift from $n = N + 1$ changes only the band endpoints. Thus the same upper bound applies to the indices satisfying the three conditions of Theorem 6.1. This is an ordinary consequence of the cited sieve argument, not a new lower bound or an asymptotic formula for the number of successful indices.

The observed proportions in Section 9 decrease across the sampled bands. These finite observations establish neither an asymptotic rate nor the required existence beyond every cutoff. Counting by prime size and counting by gap index also give different cutoffs and must not be compared without converting between them.

8.4 Recurring gap values differing by two do not suffice

Theorem 6.4 needs values differing by 2 at a prescribed distance h and a complete tail difference in a signed interval. Recurrence of two values in a residue class supplies neither requirement. We demonstrate this with an integer sequence whose dyadic sum is rational and whose complete tail differences are integral, while retaining the recurrence and growth properties stated below. This is not a sequence of actual prime gaps.

We reverse the usual order: choose integer U_n first, then set $a_n = 2U_{n-1} - U_n$. The construction must still make the coefficients positive and ensure $2^{-n}U_n \to 0$, so that these are actual tails. Constant tails $U_n = 4$ give constant coefficients $a_n = 4$. Raising one isolated tail to 6 changes the two affected coefficients from 4, 4 to 2, 8, with the same combined dyadic contribution. A slowly growing even background will supply cumulative growth $n \log n$; factorial indices will impose the recurrence in residue classes while keeping these changes sparse.

Theorem 8.8 (recurring values are not enough)**.** *There is a sequence $(a_n)_{n\ge1}$ of positive even integers with the following properties. The values 2 and 4 each occur infinitely often at indices divisible by every fixed $t \ge 1$. The sequence is unbounded, not eventually periodic, and satisfies $a_n = O(\log n)$. The series $\sum_{n\ge1} a_n 2^{-n}$ equals 6, and every scaled tail $\sum_{j\ge1} a_{N+j}2^{-j}$ is an integer, so every tail shift is integral. The increasing odd sequence $P_n = 3 + \sum_{j\le n} a_j$ satisfies $P_n \sim n \log n$.*

Proof. Put $b_n = 2\lceil \frac{1}{2}\log(n+64)\rceil$, so that b_n is even, $b_n \ge 6$, and $b_n = \log(n+64) + O(1)$. Since $\frac{1}{2}\log(n+65) - \frac{1}{2}\log(n+63) < 1$ for every $n \ge 1$, consecutive increments of b lie in $\{0, 2\}$ and $b_{n+1} - b_{n-1} \le 2$. Call an index n *special* when $n = k!$ or $n = 2\,k!$ for some $k \ge 5$, and define

$$U_n = \begin{cases} 2b_{n-1} - 2, & n = k!,\ k \ge 5, \\ 2b_{n-1} - 4, & n = 2\,k!,\ k \ge 5, \\ b_n, & \text{otherwise}, \end{cases} \qquad a_n = 2U_{n-1} - U_n \quad (n \ge 1).$$

For $k \ge 5$ one has $k! < 2\,k! < (k+1)!$, so all special indices are distinct. They are even and therefore never adjacent. Each U_n is an even integer by construction, so each a_n is an even integer.

At an ordinary index whose predecessor is also ordinary, $a_n = 2b_{n-1} - b_n = b_{n-1} - (b_n - b_{n-1}) \geq 6 - 2 = 4$. At a special index the predecessor is ordinary and $a_n = 2b_{n-1} - (2b_{n-1} - r) = r$, which is 2 at $n = k!$ and 4 at $n = 2\,k!$. Immediately after a special index carrying r,

$$a_{n+1} = 2(2b_{n-1} - r) - b_{n+1} \geq 4b_{n-1} - 2r - (b_{n-1} + 2) = 3b_{n-1} - 2r - 2 \geq 8.$$

Every a_n is therefore a positive even integer, and $U_n = O(\log n)$ gives $a_n = O(\log n)$. For $k \geq 5$, the index $n = k! + 2$ and its predecessor are ordinary: both lie strictly between $k!$ and $2\,k!$, where there are no special indices. Along these indices $a_n \geq b_{n-1} - 2 \to \infty$. Thus (a_n) is unbounded and hence not eventually periodic.

The definition $a_n = 2U_{n-1} - U_n$ is exactly $a_n 2^{-n} = U_{n-1}2^{-(n-1)} - U_n 2^{-n}$, so for every $N \geq 0$ and $m \geq 1$

$$\sum_{j=1}^{m} \frac{a_{N+j}}{2^j} = U_N - \frac{U_{N+m}}{2^m}.$$

Since $U_n = O(\log n)$ the endpoint tends to zero, so the tail at N equals the integer U_N; at $N = 0$ the series equals $U_0 = b_0 = 6$. Every difference $U_{N+h} - U_N$ is an integer, so every tail shift is integral.

For each t and every sufficiently large k we have $t \mid k!$, hence $t \mid 2\,k!$, and $a_{k!} = 2$ while $a_{2k!} = 4$: both values recur infinitely often at indices congruent to 0 modulo t.

Finally, summing $a_j = 2U_{j-1} - U_j$ gives $\sum_{j=1}^{n} a_j = 2U_0 + \sum_{j=1}^{n-1} U_j - U_n$. The baseline $\sum_{j<n} b_j = n \log n + O(n)$. The special indices up to n number $O(\log n / \log\log n)$ and each changes the summand by $O(\log n)$, so their total contribution is $O((\log n)^2) = o(n)$. Hence $P_n = 3 + \sum_{j \leq n} a_j \sim n \log n$, and P_n is odd and increasing. □

The factorial indices used above give recurrence in the zero residue class. The following separate construction strengthens this to every residue class.

8.5 A rational series with recurring values in every residue class

There is a synthetic integer sequence $(a_n)_{n \geq 1}$ with all of the following properties. Each a_n is positive and divisible by 2, and

$$a_n \leq 4 \log(n + 1) + 24.$$

For every $t \geq 1$, every $0 \leq r < t$, and every cutoff N, there are $i, j \geq N$ such that

$$i \equiv j \equiv r \pmod{t}, \qquad a_i = 2, \quad a_j = 4.$$

For every integer B and cutoff N, some $n \geq N$ satisfies $a_n > B$. For every $h \geq 1$ there is no cutoff after which $a_{n+h} = a_n$ for all n. Nevertheless, all the complete tails

$$U_N = \sum_{j \geq 1} \frac{a_{N+j}}{2^j} \quad (N \geq 0)$$

are integers, $U_0 = 6$, and $U_{N+h} - U_N \in \mathbb{Z}$ for every $N, h \geq 0$. The sequence

$$P_N = 3 + \sum_{j=1}^{N} a_j$$

is strictly increasing and odd, with $P_N/(N\log N) \to 1$. These are synthetic positions; no P_N is asserted to be prime.

Enumerate the triples (t, r, m) with $t \geq 1$, $0 \leq r < t$ and $m \geq 0$, each once. The extra index m repeats each residue-class requirement infinitely often; its parity will select the value 2 or 4. At the kth step choose a centre $c_k \equiv r \pmod{t}$, subject to $c_0 \geq 100$, $c_{k+1} \geq c_k + 3$ and $c_k \geq 2^{k^2}$ for $k \geq 1$. Each arithmetic progression is unbounded, so these lower bounds can be met recursively. Set $v_k = 2$ for even m and $v_k = 4$ for odd m. Thus each value is assigned beyond every cutoff in each residue class. Use the even baseline

$$b_n = 6 + 2\left\lfloor \frac{\log(n+1)}{2} \right\rfloor, \qquad U_n = \begin{cases} 2b_{n-1} - v_k, & n = c_k, \\ b_n, & n \notin \{c_k : k \geq 0\}, \end{cases}$$

and define $a_n = 2U_{n-1} - U_n$. The separation of the centres will ensure that $a_{c_k} = v_k$.

The baseline is even and at least 6. For $n \geq 1$, the increase of $\log(n+1)$ over two consecutive steps is less than 2, so the floor in the definition gives $b_n - b_{n-1} \in \{0, 2\}$ and $b_{n+1} - b_{n-1} \leq 2$. Away from a centre and its successor, $a_n = 2b_{n-1} - b_n \geq 4$. At a centre, $a_{c_k} = v_k$. Immediately afterwards,

$$a_{c_k+1} = 4b_{c_k-1} - 2v_k - b_{c_k+1} \geq 3b_{c_k-1} - 2v_k - 2 \geq 8.$$

Since the centres are at least three apart, these cases exhaust all indices. Also $0 < U_n \leq 2b_n$, so $a_n \leq 4b_{n-1} \leq 4\log(n+1) + 24$. The indices $n = c_k + 2$ lie outside the centres and their successors, because consecutive centres are at least three apart. Along these indices $a_n \geq b_{n-1} - 2 \to \infty$. Thus the sequence is unbounded and cannot be eventually periodic.

Finite telescoping gives

$$\sum_{j=1}^{m} a_{N+j} 2^{-j} = U_N - U_{N+m} 2^{-m}.$$

The bound $U_n = O(\log(n+1))$ makes the last term tend to zero. Hence the complete tail equals the integer U_N, and its initial value is 6. For the cumulative growth, $b_n = \log(n+1) + O(1)$ and there are at most $1 + \sqrt{\log_2 N}$ centres up to N. Replacing b_n by U_n changes its partial sum by $O((\log N)^{3/2})$. Therefore

$$\sum_{j=1}^{N} a_j = 2U_0 + \sum_{j=1}^{N-1} U_j - U_N = N\log N + O(N),$$

which gives the stated asymptotic for P_N.

The full statement is kernel-checked in Lean.

Even with all the displayed growth and recurrence properties, every tail difference is integral. Requiring the values 2 and 4 in every residue class therefore does not supply the signed tail interval. The construction is not a prime-gap sequence: its logarithmic bound excludes the extreme large gaps of the primes, so hypotheses using those gaps are not tested by this example.

8.6 Polynomial coefficients and telescoping sums

We can construct further rational sums by starting with a polynomial U_n and defining $c_{n+1} = 2U_n - U_{n+1}$. Polynomial growth gives the vanishing dyadic boundary condition, so the recurrence identifies U_n as the complete tail. A quadratic choice already makes the coefficients positive and increasing.

Proposition 8.9 (a quadratic sequence with integral tails)**.** *Put $c_n = 2(n^2 + 4n + 2)$ and $U_n = 2(n + 4)^2$. Then c_n is positive, even and strictly increasing, $U_{n+1} = 2U_n - c_{n+1}$, every shift $U_{N+h} - U_N$ is integral, $c_{n+1} - c_n = 4n + 10$ is never ± 2, and*

$$\sum_{j\geq 1} \frac{c_j}{2^j} = 32.$$

Proof. Direct expansion gives the recurrence, and the finite telescope is $\sum_{j=1}^{n} c_j 2^{-j} = 32 - 2(n + 4)^2 2^{-n}$, whose last term tends to zero. The remaining assertions follow from the integer values of U_n and from $c_{n+1} - c_n = 4n + 10 \geq 10$. □

The series starts at $j = 1$ and equals the initial rescaled tail $U_0 = 32$. Under the opening convention its sum is $\sum_{n\geq 0} c_n 2^{-(n+1)} = (c_0 + 32)/2 = 18$. Positivity, parity, strict growth, unboundedness and nonperiodicity are therefore jointly compatible with rationality, and the adjacent-gap condition of Theorem 6.4 fails at every index. The telescoping mechanism is the one used in the note of ChatGPT 5.4 Pro (orchestrated by Vjeko Kovač) against the variable-denominator expectation of Erdős [24, Theorem 1 and proof, pp. 1–2]; the sequence is not a result about Problem #251.

Rationality also fails to force the coefficients to repeat. Let $K : \mathbb{N} \to \mathbb{Q}$ be arbitrary and put $\kappa_n = 2K_n - K_{n+1}$: the coefficient of the telescoping series.

Proposition 8.10 (exact telescoping)**.** *For every $n \geq 0$, $\sum_{i=0}^{n-1} \kappa_i 2^{-(i+1)} = K_0 - K_n 2^{-n}$.*

Proof. The sum is empty when $n = 0$. Passing from n to $n+1$ adds $(2K_n - K_{n+1})2^{-(n+1)} = K_n 2^{-n} - K_{n+1} 2^{-(n+1)}$, which replaces the terminal term by the required one. □

The identity $\kappa_n = 2K_n - K_{n+1}$ has the algebraic form used in the rationality criterion of Erdős and Straus when $a_n = 2$. That criterion has additional hypotheses: for integers b_n and positive integers a_n with $a_n > 1$ for all large n and $|b_n|/(a_{n-1}a_n) \to 0$, the series $\sum_n b_n/(a_1 \cdots a_n)$ is rational exactly when some positive integer B and integers c_n satisfy $Bb_n = c_n a_n - c_{n+1}$ and $|c_{n+1}| < a_n/2$ for all large n [7, Theorem 2.1, pp. 85–86]. With $a_n = 2$, the smallness hypothesis becomes $|b_n|/4 \to 0$. Since the b_n are integers, this forces $b_n = 0$ eventually. Thus the criterion in this specialisation does not apply even to a bounded integer sequence that is nonzero infinitely often, let alone to the positive prime gaps. Proposition 8.10, by contrast, imposes no growth condition on K. The infinite series converges precisely when $2^{-n}K_n$ has a finite limit, and then its sum is $K_0 - \lim_{n\to\infty} 2^{-n}K_n$. The sum equals K_0 precisely when that limit is zero; polynomial growth suffices. For $K_n = 2^n$, every κ_n is zero, so the series converges to 0, not to $K_0 = 1$.
Taking $K_0 = \frac{5}{2}$ and $K_n = 2n + 2$ for $n \geq 1$ gives $\kappa_0 = 1$ and $\kappa_n = 2n$, so the coefficients $1, 2, 4, 6, 8, \ldots$ are positive, even after the first term, unbounded and not eventually periodic, while their dyadic sum is $\frac{5}{2}$. Rationality alone therefore cannot imply eventual

periodicity even for a positive, parity-correct integer coefficient sequence, and rationality cannot be contradicted merely by Proposition 6.3.

The arithmetic-progression formulations of nonintegrality do not create new prime-distribution information. Their full definitions and exact hypotheses are preserved in Appendix E.

9 Finite numerical searches

An enclosure must control both the omitted tail and arithmetic error. The scans supply no bound for accumulated double-precision rounding, so their counts remain observations even when the omitted tail is rigorously bounded. By contrast, Proposition 6.6 and Section 7 use exact integer comparisons with linked Lean checks.

Over the 6 841 648 primes below 1.2×10^8, the scan tests $h = 1, \dots, 16$ separately. For each offset it records indices satisfying the signed inequalities in Theorem 6.4, with observed proportions between 0.00418 and 0.008248 and nearly balanced signs. The reported counts do not require one index to work for all offsets. At $h = 1$ it records 56 427 hits among 6 841 564 candidate indices, the last at the prime 119 995 753. Multiplying the observed proportion in each band by that band's mean of $\log p$ gives 0.17671 in the first band and 0.13148 in the last. This finite observation establishes no asymptotic counting law and no occurrence beyond every cutoff. The search output is public.

A second scan uses the 1 270 607 primes below 2×10^7 and pairs of indices in the half-open range $1\,143\,545 \le n, m < 1\,270\,540$. It searches each residue class modulo each $1 \le t \le 20$ for the condition of Theorem 5.5, selecting pairs with a next-gap difference of 2 and a tail difference in the specified interval. At $t = 20$, every residue class has at least 90 150 counted pairs; pairs may share indices. For each class, the program also records the larger index of a selected late witness. The minimum of those recorded indices is 1 270 520; it need not come from the class with the fewest pairs. For the saved sample pairs, floating-point arithmetic gives both small-difference inequalities. The scan supplies neither arbitrary moduli nor witnesses beyond every prescribed cutoff. Its search output is public. The computation guide gives the programs, dependencies and commands to rerun both searches and the 80000-bit continued-fraction calculation of Section 7. Appendix D.7 corrects the stated justification of the tail bound in the free-pair program.

10 Remaining prime-gap estimates

For the actual tails, Theorem 5.3 makes the first question equivalent to irrationality. The second requires two small differences and a gap mismatch at the same index. A finite certificate verifies one instance; both questions require witnesses arbitrarily late for every positive shift.

Problem 10.1 (nonintegral tail differences of each positive length)**.** For every $h \ge 1$ and every N_0, prove that some $N \ge N_0$ satisfies

$$\sum_{j \ge 1} \frac{g_{N+h+j} - g_{N+j}}{2^j} \notin \mathbb{Z}. \tag{9}$$

Problem 10.2 (two small tail differences at arbitrarily large indices)**.** For every $h \geq 1$ and every N_0, prove that some $N \geq N_0$ satisfies

$$|\sigma_h(N)| < 1, \qquad |\sigma_h(N+1)| < 1, \qquad g_{N+h+1} \neq g_{N+1}. \tag{10}$$

Problem 10.1, like the congruent-index criterion of Theorem 5.5, is exact; Problem 10.2 is only sufficient. The counterexamples in Section 8 show why the retained coefficient data do not supply the missing tail information. They are not counterexamples about actual primes.

The successful indices have density zero by Theorem 8.7, so the required existence theorem must detect a sparse set. A finite prefix is compatible with a rational comparison series. Supplemented by a proved remainder bound, however, it can certify one actual pair, as in Proposition 6.6; it does not establish occurrence beyond every cutoff.

Proposition 6.5 gives sufficient integer inequalities ensuring $|\sigma_h(N)| < 1$ at a chosen truncation length L. Failure of such a test need not mean that the complete tail lies outside the required interval: its error interval may still cross an endpoint. For fixed h and $\varepsilon > 0$, the choice $L = \lceil (4+\varepsilon)\log_2(N+2) \rceil$ gives $E_{h,N,L} = O_{h,\varepsilon}(N^{-\varepsilon})$. Indeed, $L = O_\varepsilon(\log N)$, both arguments of the quartic P are $O_{h,\varepsilon}(N)$, and $2^{-L} \leq (N+2)^{-4-\varepsilon}$. Requiring a fixed positive margin from both endpoints is stronger than the strict inequalities in (10). At this depth, the test bounds the signed integer $2^L F_{h,N,L}$, not just its residue modulo 2^L, and also requires $g_{N+h+1} - g_{N+1} = \pm 2$. For fixed h and ε, checking one certificate uses $O_\varepsilon(\log N)$ gap values and the explicit remainder bound. This counts data, not bit operations. The unproved input is the existence of certified witnesses beyond every cutoff for each fixed h; constants uniform in h are not required.

The following published results control gap sizes and clusters, not the signed weighted sums required by this criterion. Zhang's bounded-gap theorem [21, Theorem 1, p. 1122] produces infinitely many bounded consecutive-prime gaps. Maynard bounds $\liminf_n (p_{n+m} - p_n)$ for every fixed m, giving bounded clusters of every fixed size [16, Theorem 1.1, p. 384], with the explicit unconditional bound $\liminf_n (p_{n+1} - p_n) \leq 600$ [16, Theorem 1.3, p. 385]. Polymath subsequently improved this to 246 [35, Theorem 1.4(i) of arXiv v4]; neither numerical bound controls the signed weighted tails required here. The large-gap theorem of Ford, Green, Konyagin, Maynard and Tao gives an effective lower bound for the largest single consecutive-prime gap below X [8, Theorem 1]. A theorem giving bounded clusters at each fixed cluster size does not by itself supply a weighted condition on blocks whose length grows with the basepoint, and none of these results is used as a proof input here.

Conditionally the picture is different. Tao's comment names uniform quantitative prime-tuples control of about $\log\log n$ consecutive gaps as a possible approach [23, comment of 7 October 2025], and Land's draft carries that out under Kuperberg's uniform prime-tuples conjecture [13]. The counterexamples in Section 8 show that the listed size, congruence and block-statistical properties alone do not force irrationality. These constructions are not shown to enumerate the consecutive primes, so they do not rule out arguments that combine those properties with further information about primes.

Funding and competing interests. This work received no external funding. The author declares no competing interests.

Acknowledgements. The problem numbering follows the Erdős Problems catalogue maintained by Thomas Bloom [22]. The logarithmic-scale construction and the transfer of nonconcentration come from earlier working materials of this project, which were not an independent mathematical review.

A An elementary polynomial bound for the primes

We derive the bound $p_n \le 1250(n+1)^4$ by comparing the size and prime factorisation of a central binomial coefficient, following Erdős's proof of Chebyshev's theorem [5, §1, pp. 194–196]. With $\pi(x)$ denoting the number of primes at most x, we first prove, for $m \ge 4$,

$$4^m < m\binom{2m}{m} \le m(2m)^{\pi(2m)}. \tag{11}$$

For the first inequality, $m\binom{2m}{m}/4^m$ equals $70/64$ at $m = 4$ and its ratio at successive indices is $(2m+1)/(2m) > 1$. For the second, the exponent of a prime ℓ in $\binom{2m}{m}$ is $\sum_{k\ge1}(\lfloor 2m/\ell^k\rfloor - 2\lfloor m/\ell^k\rfloor)$, each summand is 0 or 1, and every summand with $\ell^k > 2m$ vanishes; the exponent is therefore at most $\lfloor \log_\ell(2m)\rfloor$. Thus the full power of each prime appearing in $\binom{2m}{m}$ is at most $2m$. Multiplying over at most $\pi(2m)$ distinct primes gives the second inequality.

Now fix $n \ge 0$, put $x = n+5$ and $m = x^4 \ge 625$, and suppose $\pi(2m) \le n$. Since $x \le 2^x$ and $n + 4(n+1)x = 4x^2 - 15x - 5 \le 2x^4$,

$$m(2m)^n = 2^n x^{4(n+1)} \le 2^{\,n+4(n+1)x} \le 2^{\,2x^4} = 4^m,$$

contradicting (11). Hence $\pi(2m) > n$, so at least $n+1$ primes lie below $2m$ and therefore

$$p_n \le 2(n+5)^4 \le 1250(n+1)^4,$$

the last step because $(n+5) \le 5(n+1)$ for $n \ge 0$. The coarser bound $p_n \le 1250(n+1)^4$ is checked by the kernel at line 360.

B Integer certificates

This calculation checks two different finite conclusions from freshly computed primes below 10^4: the two small-pair rows of Proposition 6.6, and the determinant and endpoint comparisons in Theorem 7.1. All arithmetic is integral; the four bracketing integers come from the linked Lean proof. The program checks those integers, not the analytic prime bound used to turn them into enclosures. Adjacent quoted strings are concatenated by Python to keep the printed integers legible.

```
from math import isqrt

primes = [n for n in range(2, 10000)
          if all(n % d for d in range(2, isqrt(n) + 1))]
gaps = [q - p for p, q in zip(primes, primes[1:])]
P = lambda x: x**4 + 8*x**3 + 36*x*x + 104*x + 150
Q = 2**40
expected = [(-662838684750, 11764181250),
```

```
            (873345886050, 12805761250)]
for N, target in zip((2, 3), expected):
    D = sum((gaps[N+1+j] - gaps[N+j])*2**(40-j)
            for j in range(1, 41))
    B = 1250*(P(N+43) + P(N+42))
    distance = min(D % Q, Q - D % Q)
    if (D, B) != target or not (abs(D)+B < Q and B < distance):
        raise ArithmeticError("tail certificate failed")
if gaps[4] == gaps[3]:
    raise ArithmeticError("gap mismatch failed")

u = int(
    "80656418571526529321760196321868980032711628291714663348 27"
    "30836077944152787174450335094078559889033699885255507461 59"
    "73558897922500842023448210201391609566636587897181681526 62"
    "0217"
)

v = int(
    "21949451244136632321439709245412633124220695246356153605 18"
    "42470773519582218168307201892899048316629550843926902486 83"
    "12162917239885377332351730406072544968385302138677814423 35"
    "1745"
)

up = int(
    "65394371014981626268824118924709052221082600085685554459 75"
    "30261633155217846686899097127598765624846200590981384174 69"
    "52328398881853163742723332773896114831173340004938675849 23"
    "912"
)

vp = int(
    "17796110757898665511984272416217096301201363232815981766 37"
    "84906052492297355019687649040361529707294916566508739385 80"
    "62030254663133898102912437860054991645374993836120818051 60"
    "767"
)

c = len(primes)
A = sum(p*2**(c-1-i) for i, p in enumerate(primes))
checks = (c == 1229,
          u > 0 and v > 0 and up > 0 and vp > 0,
          up*v - u*vp == 1,
          u*2**c < A*v,
          (2*A + 5000*(c+1)**4)*vp < up*2**(c+1),
          v+vp >= 2**589,
          2**589 > 10**177)
if not all(checks):
    raise ArithmeticError("denominator certificate failed")
print("Both tail rows and the denominator floor verified.")
```

C Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The source links identify definitions and proof steps at fixed revisions; the short paper's source index retains its different revision in Appendix G. The formal sparse construction changes one coordinate per stage, whereas the printed proof uses triples. Both fix a set containing all correction supports, the interval of sums and the modulus cutoffs before choosing a value. The formal proofs use the published inputs stated below; the linked declarations do not check the different printed arguments step by step.

Lean 4 [14] and mathlib [15] supply the proof-checking environment. The evidence record lists the declarations and checks. The programs for the searches in Section 9 are in the computation guide. The program in Appendix B checks the adjacent pair and the bound 2^{589}; the independent program of Section 7 checks the certificate of Theorem 7.2. Formal checking does not establish novelty or attribution.

The summation-by-parts identity and the polynomial prime bound concern the actual primes. The recurrence results apply to arbitrary integer coefficients and, as stated, to rational or real sequences; substituting the actual prime-gap tail requires its convergence and boundary condition. The finite certificates use the same polynomial bound to control the omitted terms. The counterexamples concern other coefficient sequences and construct no new primes.

The Lean proofs of three results assume published inputs that the Lean development does not prove. The Lean proof of Theorem 8.7 assumes Schlage-Puchta's Lemma 4 [18, Lemma 4, pp. 5–6]; for a fixed nonzero integer polynomial in a fixed block of consecutive prime gaps, it states that the set of indices where the polynomial vanishes has density zero. Those of Corollaries 8.4 and 8.6 assume that lemma and the prime number theorem in the form $p_n \sim n \log n$ [17, Chapter 6]. What each statement adds to these inputs is checked, including the estimate that the corrected sequence of Corollary 8.6 differs from the prime gaps at $O_\varepsilon(n/\log\log n)$ indices below n, which uses no information about the distribution of primes.

D Further criteria, examples and computational details

These further results concern the recurrence and actual-tail tests, not the sparse construction. We separate exact reformulations from stronger sufficient conditions, explain why refining one fixed enclosure differs from controlling enclosures at changing indices, and compare repeated tail values with the nonintegral differences the criteria require. A bounded counterexample, the complete numerical table and two corrections are retained below.

D.1 Totient-length shifts and persistence of integrality

Once the denominator is odd, Euler's congruence specifies an integral shift of totient length. We record this choice and then prove that an integral shift persists under each subsequent recurrence step.

Proposition D.1 (a shift of totient length)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer coefficients. If the reduced denominator d of T_N is odd, then $\sigma_{\varphi(d)}(N)$ is an integer.*

Proof. Since d is odd, 2 and d are coprime, so Euler's congruence gives $d \mid 2^{\varphi(d)} - 1$. Writing $2^{\varphi(d)} - 1 = dk$ and $T_N = u/d$ in lowest terms, $(2^{\varphi(d)} - 1)T_N = ku$ is an integer, and the integral-shift criterion transfers this to the shift. □

Proposition D.2 (propagation)**.** *Let $T : \mathbb{N} \to \mathbb{R}$ satisfy $T_{n+1} = 2T_n - a_{n+1}$ with integer coefficients, and define $\sigma_h(N) = T_{N+h} - T_N$. For fixed $h, N \geq 0$, if $\sigma_h(N)$ is an integer, then $\sigma_h(N + k)$ is an integer for every $k \geq 0$.*

Proof. The shift step identity gives $\sigma_h(N + 1) = 2\sigma_h(N) - (a_{N+h+1} - a_{N+1})$, an integer combination of an integer and two coefficients. Induct on k. □

These are the totient shift and the propagation theorems cited in the note. For example, if $\operatorname{den} T_N = 3$ then $\varphi(3) = 2$ and $3T_N$ is an integer, so $\sigma_2(N)$ is integral while $\sigma_1(N)$ is not; if $\operatorname{den} T_N = 5$ then $\sigma_4(N)$ is integral. In the orbit with every coefficient zero and $T_0 = 1/12$, the denominator is $12 = 2^2 \cdot 3$, the orbit reaches $T_2 = 1/3$ with odd denominator at $s = 2$, and $h = \varphi(3) = 2$ is exactly the shift length seen to be integral from index 2 onwards. The totient supplies one length, while the exact criterion is $\operatorname{den}(T_N) \mid 2^h - 1$. At an index with odd denominator $d > 1$, the least positive length is the multiplicative order of 2 modulo d. When $d = 1$, every positive length works; before the denominator becomes odd, no positive length works.

D.2 The finite truncation criterion

For the actual gaps, we enclose each complete weighted difference by a finite sum and a bound on the omitted terms. A nonnegative summable majorant gives such an enclosure, but need not provide a computable remainder. Proposition 6.5 supplies the rational bound used in the finite tests.

Proposition D.3 (finite truncation)**.** *Let $M : \mathbb{N} \to \mathbb{R}$ satisfy $M(n) \geq g_n$ for every n and $\sum_{n \geq 0} M(n)2^{-n} < \infty$, and put*

$$S_{h,N,L} = \sum_{j=1}^{L} \frac{g_{N+h+j} - g_{N+j}}{2^j}, \qquad R_{h,N,L}(M) = \sum_{j>L} \frac{M(N+h+j) + M(N+j)}{2^j}.$$

If for every $h \geq 1$ and every N_0 there are $N \geq N_0$ and $L \geq 1$ with $\operatorname{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M)$, then the nonintegrality condition in Problem 10.1 holds.

Proof. The part of $\sum_{j \geq 1}(g_{N+h+j} - g_{N+j})2^{-j}$ omitted from $S_{h,N,L}$ has absolute value at most $R_{h,N,L}(M)$, so under the displayed inequality the full sum lies at positive distance from every integer. □

With $M(n) = 1250(n+2)^4$, the quantities $S_{h,N,L}$ and $R_{h,N,L}(M)$ are exactly $F_{h,N,L}$ and $E_{h,N,L}$ from Proposition 6.5. The same enclosure therefore tests separation from the integers, without requiring either of the two consecutive differences to be small. To compute the distance from the truncated sum to $\mathbb{Z}$, write the weighted block sum as

$$D_{h,N,L} = \sum_{j=1}^{L} 2^{L-j}(g_{N+h+j} - g_{N+j}), \qquad S_{h,N,L} = \frac{D_{h,N,L}}{2^L}.$$

Then

$$\operatorname{dist}(S_{h,N,L}, \mathbb{Z}) = 2^{-L} \min\{D_{h,N,L} \bmod 2^L,\ 2^L - (D_{h,N,L} \bmod 2^L)\},$$

where $D_{h,N,L} \bmod 2^L$ is the least nonnegative residue, including when $D_{h,N,L} < 0$. The criterion requires this residue to be more than $2^L R_{h,N,L}(M)$ from both 0 and 2^L. One successful comparison certifies nonintegrality at that index; a failed comparison is inconclusive. Irrationality requires such certificates arbitrarily late for every positive h. At prescribed logarithmic depth, the needed residue must be farther from both endpoints than the error bound. With the polynomial majorant in Proposition 6.5, the remainder bound is rational and explicit, so the whole test reduces to integer comparisons. The modular rewriting is an ordinary deduction, not a separately checked Lean declaration.

Proposition D.4 (completeness of finite separation)**.** *Suppose $D \in \mathbb{R}$, $S_L \in \mathbb{R}$ and $R_L \geq 0$ satisfy $|D - S_L| \leq R_L$ and $R_L \to 0$. Then*

$$D \notin \mathbb{Z} \quad \Longleftrightarrow \quad \text{there exists } L \text{ with } \operatorname{dist}(S_L, \mathbb{Z}) > R_L.$$

Proof. The reverse implication follows from the triangle inequality. For the forward implication put $\delta = \operatorname{dist}(D, \mathbb{Z}) > 0$. For all sufficiently large L, one has $R_L < \delta/2$, and the 1-Lipschitz property of distance gives $\operatorname{dist}(S_L, \mathbb{Z}) \geq \delta - R_L > R_L$. No monotonicity of the errors is required. □

For fixed h, N, increasing the depth eventually certifies every nonintegral $\sigma_h(N)$. This is not a decision procedure for integrality: finitely many failed tests prove nothing, although a proof of failure at every depth implies integrality. It gives neither a prescribed depth as N varies nor arbitrarily late successful indices. For example, $[1 - 2^{-N}, 1 + 2^{-N}]$ contains both the noninteger $1 + 2^{-2N}$ and the integer 1 for every $N \geq 1$. The radii shrink, but the values also change; no enclosure certifies nonintegrality. Completeness keeps the value fixed while refining its enclosure.

D.3 Nonintegral differences at multiples of each shift

We may ask for a nonintegral multiple of each prescribed shift length. The following formulation still has the full strength of irrationality, as the argument after the question shows.

Problem D.5 (multiples of each shift)**.** For every $r \geq 1$, is there an $m \geq 1$ such that the tail differences of length mr are nonintegral at arbitrarily large indices, as in (9)?

This is an exact reformulation, not a weaker mathematical target. If T_0 is irrational, every positive shift is nonintegral by the block identity, so the stated condition follows with $m = 1$. Conversely, rationality gives a shift $h \geq 1$ that is integral at every sufficiently late index. For each positive integer m, the identity

$$T_{N+mh} - T_N = \sum_{j=0}^{m-1} (T_{N+(j+1)h} - T_{N+jh})$$

makes every multiple mh eventually integral. Taking $r = h$ contradicts the stated condition. This ordinary deduction uses the checked rationality classification and a finite telescope; rearranging the quantifiers supplies no new estimate for the actual prime gaps.

D.4 Repeated tail values and congruent indices

For the complete actual tails $T_N = \sum_{j\geq 1} g_{N+j} 2^{-j}$, the prime number theorem controls the mean size without assuming rationality. Rationality would then restrict these tails to a fixed rational lattice, forcing many repetitions by pigeonhole. Repetition gives difference zero, not the nonintegral difference required at congruent indices. We make both steps explicit. These are ordinary deductions; the linked formal results establish the recurrence and the congruent-index criterion.

Repeated tail values. For an integer $X \geq 0$, summing $g_{N+1} = 2T_N - T_{N+1}$ over $0 \leq N < X$ gives the exact identity

$$\sum_{N=0}^{X} T_N = p_{X+1} - p_1 - T_0 + 2T_X \leq (p_{X+1} - p_0) + 2T_X.$$

The inequality uses $T_0 \geq 0$ and $p_1 \geq p_0$. The prime number theorem [17, Ch. 6] also gives

$$\frac{T_X}{p_X} = \sum_{j\geq 1} \frac{g_{X+j}}{2^j p_X} \longrightarrow 0. \tag{12}$$

Indeed, for each fixed j, both p_{X+j+1}/p_X and p_{X+j}/p_X tend to 1, so their difference g_{X+j}/p_X tends to zero. For the uniform bound needed to pass the limit through the sum, the two-sided estimates $p_n \asymp n \log n$ give one constant K such that, for $X \geq 2$ and $j \geq 1$,

$$\begin{aligned} \frac{g_{X+j}}{p_X} &\leq K \frac{(X+j+1)\log(X+j+1)}{X \log X} \\ &\leq K(1+j)\left(1 + \frac{\log(1+j)}{\log X}\right) \leq K(1+j)^2. \end{aligned}$$

The second line uses $X + j + 1 \leq X(1 + j)$ and $\log(1 + j) \leq j \log 2 \leq j \log X$. Since $\sum_{j\geq 1} (1+j)^2 2^{-j} < \infty$, dominated convergence applies. Subtracting the exact summation identities at $2X$ and X now gives

$$\sum_{X<N\leq 2X} T_N = p_{2X+1} - p_{X+1} + 2(T_{2X} - T_X) \sim X \log X.$$

Indeed, the prime difference is asymptotic to $X \log X$, and (12) makes both tail terms $o(X \log X)$. Thus the mean complete tail in this band is asymptotic to $\log X$, without a rationality assumption. This is a mean, not a pointwise estimate. For the counting argument we only need the upper bound $\sum_{X<N\le 2X} T_N \le K_0 X \log X$, valid for any fixed $K_0 > 1$ and all sufficiently large X. For each fixed $C > 1$, Markov's inequality then leaves at least $(1 - 1/C)X$ indices with $T_N \le K_0 C \log X$.

Under rationality, take X beyond the point where the reduced denominator is the fixed odd integer d. Among the indices just selected by Markov's inequality, the values lie in $d^{-1}\mathbb{Z}_{\ge 0} \cap [0, K_0 C \log X]$. There are at most $dK_0 C \log X + 1$ such values. Pigeonhole therefore gives one value occurring at least

$$\frac{(1 - 1/C)X}{dK_0 C \log X + 1}$$

times. Taking, for example, $K_0 = 2$ gives a lower bound of order $\gg_{C,d} X/\log X$. The value may depend on X; no single value recurring in every band is established.

The growth input is essential: the rational quadratic example in Proposition 8.9 has strictly increasing tails $T_N = 2(N + 4)^2$. For the primes, however, repeated values give zero difference, not the required nonintegral one.

The terminal term in the exact identity cannot simply be omitted. The factorial-gap argument in the proof of Proposition 6.3 shows that $T_X \ge g_{X+1}/2$ is unbounded, whereas $p_1 + T_0$ is fixed. The exact identity therefore gives $\sum_{N=0}^{X} T_N > p_{X+1}$ for arbitrarily large X. It does not establish that this strict inequality holds eventually.

The interval condition at congruent indices. Suppose the sum is rational, let d be the eventual odd reduced denominator, and let t be the order of 2 modulo d, with $t = 1$ when $d = 1$. Beyond the denominator cutoff, $M \equiv N \pmod t$ forces $T_M - T_N \in \mathbb{Z}$. Neither this difference nor its negative can then lie in $(\frac{1}{2}, 1)$. Thus a witness in that interval, at two such congruent indices, would contradict rationality without a further condition on the intervening gaps. This uses only the interval occurring in Theorem 6.4; it does not assert that its other hypotheses hold for arbitrary congruent pairs.

D.5 A bounded sequence with recurring values and a rational sum

The following bounded example isolates the recurring-values argument without a growth estimate. Logarithmic growth was needed in Theorem 8.8 to obtain cumulative size $n \log n$, but is unnecessary for the recurrence itself.

Proposition D.6 (recurring values with integral tails)**.** *Put $U_0 = 4$ and, for $n \ge 1$, $U_n = 6$ when $n = k!$ for some $k \ge 3$ and $U_n = 4$ otherwise, and set $a_n = 2U_{n-1} - U_n$ for $n \ge 1$. Then $a_n \in \{2, 4, 8\}$. For every $k \ge 3$, the value 2 occurs at index $k!$ and the value 4 at index $2\,k!$, so both recur infinitely often at indices divisible by any fixed $t \ge 1$. The series $\sum_{n\ge 1} a_n 2^{-n}$ equals 4 and every tail $\sum_{j\ge 1} a_{N+j} 2^{-j}$ equals the integer U_N.*

Proof. For $k \ge 3$ the index $k!$ is even and at least 6, and $k! - 1$ is odd, so the preceding tail is 4; hence $a_{k!} = 8 - 6 = 2$, $a_{k!+1} = 12 - 4 = 8$, and $a_n = 8 - 4 = 4$ elsewhere. For $k \ge 2$ one has $k! < 2\,k! < (k + 1)!$, so $2\,k!$ is not a factorial. Its odd predecessor $2\,k! - 1 \ge 3$ is

not a factorial either; therefore $a_{2k!} = 4$. Since $t \mid k!$ for every large k, both values recur in the residue class 0 modulo t. The identity $a_n 2^{-n} = U_{n-1}2^{-(n-1)} - U_n 2^{-n}$ telescopes to $\sum_{j=1}^{m} a_{N+j}2^{-j} = U_N - U_{N+m}2^{-m}$, and U is bounded, so the endpoint vanishes. □

The indices with coefficient 2 are precisely the factorials $k!$, $k \geq 3$. They occur infinitely often with unbounded gaps, which is impossible for a value recurring in an eventually periodic sequence. Thus even this bounded coefficient sequence is nonperiodic, despite its rational dyadic sum. Its cumulative sum has linear growth. The logarithmic construction in Theorem 8.8 adds unboundedness and cumulative growth $N \log N$ without changing the telescoping mechanism. Neither construction guarantees prime cumulative positions.

D.6 Numerical counts and proportions

The scan in Section 9 used the 6 841 648 primes below 1.2×10^8 and computed double-precision tails for offsets $h \leq 16$. We reproduce its reported counts for $h = 1, \ldots, 4$. The denominator of each proportion is the number of tested basepoints, $6\,841\,565 - h$, rather than the prime count.

h	events	proportion	$\delta = +2$	$\delta = -2$	last event at prime
1	56 427	0.008248	28 022	28 405	119 995 753
2	31 979	0.004674	15 742	16 237	119 993 807
3	30 233	0.004419	15 093	15 140	119 998 321
4	29 262	0.004277	14 606	14 656	119 993 473

These are separate counts for each offset $1 \leq h \leq 16$, not counts of indices successful for all offsets at once. The proportions range from 0.00418 to 0.008248, with nearly balanced signs at each offset. For $h = 1$, the first row below gives the proportions in the eight bands; the second multiplies each by its band's mean of $\log p$:

0.011285, 0.008951, 0.008303, 0.007936, 0.007651, 0.007507, 0.007253, 0.007094;

0.17671, 0.15058, 0.14420, 0.14067, 0.13766, 0.13667, 0.13333, 0.13148.

The ratio of the last rescaled proportion to the first is 0.744 at $h = 1$ and lies between 0.744 and 0.8121 across the measured offsets. Theorem 8.7 gives limiting density zero for each fixed offset. The bound $O_h(X/\log\log X)$ derived after its proof is only an upper bound; it predicts neither a monotone decline nor an asymptotic proportion. The finite-band ratios therefore do not establish an asymptotic rate.

D.7 Two corrections

We distinguish a bound for the actual tail from the polynomial series used to compute it, and then correct an insufficient offset condition.

The tail bound in the free-pair program. The 4096-bit free-pair program linked through Section 9 bounds T_s by summing the first 400 terms of $1250(s+j+2)^4 2^{-j}$ after rounding each down, then adding 1. This does not necessarily bound that polynomial series: at the recorded $s = 1\,270\,306$, the returned value is 40 below the exact sum $1250P(s+2)$, with P as in Proposition 6.5.

The returned bound nevertheless has a valid justification. The stronger estimate $p_n \le 2(n+5)^4$ proved in Appendix A gives

$$T_s \le 2P(s+6) < 625(s+3)^4 \qquad (s \ge 0).$$

The last expression is already the first summand of the program, and all its remaining summands are nonnegative. The strict inequality follows from

$$625(s+3)^4 - 2P(s+6) = 623s^4 + 7436s^3 + 32958s^2 + 62972s + 40437 > 0.$$

Thus the returned bound exceeds the actual tail, even though its stated comparison with the larger polynomial series fails. This validates the analytic bound only; it neither controls floating-point errors in other scans nor reruns the denominator-exclusion search.

An insufficient condition. Theorem 5.5 allows the offset to vary. For each modulus t, one must still find pairs of congruent indices beyond every prescribed cutoff whose tail difference is nonintegral. Two even coefficient values differing by 2, each occurring infinitely often in the same residue class of indices, do not give the required tail difference for an integer-coefficient recurrence: the sequence of Theorem 8.8 has both properties and only integral tail shifts.

E Arithmetic-progression reformulations of integrality

A longer observed block can appear to provide a stronger arithmetic test. Here it does not: the recurrence inserts the same block sum into the quantity tested and into the comparison progression, so that sum cancels. The moving progression tests even integrality of the original difference. Under the stated growth bound, separation from the fixed progression $2^r\mathbb{Z}$ tests nonintegrality. We record the two reductions separately.

Fix h and abbreviate $D_N = \sigma_h(N)$ and $\delta_N = g_{N+h+1} - g_{N+1}$. Iterating over a longer block gives

$$B_{h,N,r} = \sum_{i=0}^{r-1} 2^{r-1-i}\delta_{N+i}; \qquad D_{N+r} = 2^r D_N - B_{h,N,r}.$$

Both the progression $-B_{h,N,r} + 2^{r+1}\mathbb{Z}$ and the value D_{N+r} contain the same integer offset. Substituting the recurrence cancels it, leaving a condition on D_N alone. Thus increasing the block length in this particular test adds no arithmetic information.

Theorem E.1 (equivalent arithmetic-progression tests)**.** *For every rational dyadic tail recurrence and all $h, N, r \ge 0$,*

$$D_{N+r} \in -B_{h,N,r} + 2^{r+1}\mathbb{Z} \quad \Longleftrightarrow \quad D_N \in 2\mathbb{Z}. \tag{13}$$

Consequently, if every δ_N is even, then

$$\begin{aligned}(\forall N_0\ \exists N, r:\ N_0 < N \text{ and } D_{N+r} \notin -B_{h,N,r} + 2^{r+1}\mathbb{Z}) \\ \iff \quad D_N \notin \mathbb{Z} \text{ for arbitrarily large } N.\end{aligned} \tag{14}$$

There is a second equivalence. Let $b : \mathbb{N} \to \mathbb{Q}$ satisfy $|D_N| \le b(N)$ for every N, and suppose that for every N and every positive integer q there is an r with

$$2b(N+r)q < 2^r. \tag{15}$$

Then

$$\begin{aligned}\forall N_0\ \exists N, r:\ N_0 < N \text{ and } \forall z \in \mathbb{Z}, \quad b(N+r) < |B_{h,N,r} - 2^r z| \\ \iff \quad D_N \notin \mathbb{Z} \text{ for arbitrarily large } N.\end{aligned} \tag{16}$$

Proof. Substituting $D_{N+r} = 2^r D_N - B_{h,N,r}$ into the first membership condition cancels the observed block from both sides and leaves $D_N = 2z$. This proves (13). When every δ_N is even, the recurrence also gives $D_{N+1} \in 2\mathbb{Z}$ if and only if $D_N \in \mathbb{Z}$. Thus eventual integrality would force eventual even integrality and contradict the left side of (14). This proves the forward implication. Conversely, given a cutoff, choose a nonintegral D_N beyond it and take $r = 0$ in (13).

For (16), eventual integrality and the block identity give some $z \in \mathbb{Z}$ with

$$B_{h,N,r} - 2^r z = -D_{N+r},$$

so the displayed strict separation contradicts $|D_{N+r}| \le b(N+r)$. Conversely, if D_N is nonintegral, choose a positive integer q with $1/q \le \operatorname{dist}(D_N, \mathbb{Z})$. Such a q exists because $\mathbb{Z}$ is closed and $D_N \notin \mathbb{Z}$. Choose r from (15). The block identity and the triangle inequality give, for every $z \in \mathbb{Z}$,

$$|B_{h,N,r} - 2^r z| \ge 2^r |D_N - z| - |D_{N+r}| \ge \frac{2^r}{q} - b(N+r) > b(N+r).$$

The same r works for all integers z, as required. □

Evenness is essential in (14). Without it, the recurrence $T_N = N + 1$, $g_n = n - 1$ for $n \ge 1$, has $D_N = \sigma_1(N) = 1$ and $\delta_N = 1$ at every index. Thus D_N is never an even integer but is always an integer. By (13), the left side of (14) holds while the right side fails.

The growth hypothesis (15) is equivalent to

$$\liminf_{n\to\infty} 2^{-n} b(n) = 0.$$

Indeed, writing $n = N + r$, its inequality becomes $2^{-n} b(n) < 1/(2^{N+1} q)$. Since $b(n) \ge 0$, finding such an $n \ge N$ for every N and every positive integer q is exactly the stated liminf condition. In particular it holds for polynomial bounds and for Cc^n with fixed $C > 0$ and $1 < c < 2$, but fails for $b(n) = 2^n$. A limit of zero is not necessary: the bound $b(n) = 1$ at odd indices and $b(n) = 2^n$ at even indices also satisfies the hypothesis. This

example concerns the growth condition alone, not a bound for the prime tails. Only one suitable scale is needed to magnify the positive distance to $\mathbb{Z}$ beyond the error; no comparison at every sufficiently large scale is required.

The proof in fact uses no rationality of D_N: the integer q is chosen from its positive distance to $\mathbb{Z}$, not from a reduced denominator. Thus (16) holds for real integer-coefficient recurrences under the same bound and growth hypothesis. The same is true of (13), and of (14) when the coefficient differences are even. These extensions follow from the printed argument; the Lean proof covers the rational case stated in the theorem.

The first test is exactly even integrality of D_N, regardless of r. The fixed-progression test compares $2^r \operatorname{dist}(D_N, \mathbb{Z})$ with the bound on the remaining tail. When D_N is nonintegral, the growth hypothesis makes the former larger than twice the latter at a suitable scale. Rewriting the test does not provide an estimate about the distribution of consecutive primes.

E.1 A signed comparison with more analytic data

A signed variant appears in the totient companion, *The Binary Totient Series*, under "Signed interpolation with entire Dirichlet defects". We recall its different hypotheses to delimit the comparison. Around the totient sequence, signed integer changes bounded by any positive $f(n) \to \infty$ fill an interval at every reciprocal integer base. They can preserve the odd coefficients, every fixed eventual congruence, all arithmetic-progression Dirichlet singularities, and prescribed finite root-of-unity jets. The proof takes finite differences of increasing order on separated finite sets of indices. The support of the changes has upper Banach density zero. The union of the intervals spanned by those sets has ordinary density zero and upper Banach density one.

This theorem allows the bound on the corrections to grow arbitrarily slowly and retains more analytic observations. It uses both signs, however, and does not subsume the one-sided prime-gap perturbation theorem proved here. In particular, it constructs comparison integer sequences, not consecutive prime gaps, and neither theorem supplies an irrationality proof for the actual prime series. The companion contains a full ordinary proof and the precise block estimate in terms of the prescribed bound; no infinite formal theorem is claimed.

F Conclusions

The construction's fixed ordinary totals protect cumulative residues; its variable weighted totals attain the target. The resulting family contains rational and irrational sums with the stated congruences and asymptotic block statistics, but does not preserve consecutive primality.

For the actual primes, the complete tails are identified and their integral shifts give an exact criterion. The remaining task is to establish the required nonintegrality, not another recurrence identity. The small-pair route requires simultaneous witnesses arbitrarily late for every positive shift. Finite separations and denominator exclusions do not establish that infinite family of witnesses.

G Source index for the short paper

The links below identify further formal definitions and statements used in the short paper, at its original revision `99f4bf47422a` and line numbers. This local source pin applies only to the index below; Appendix C retains the long record's separate pin. Each link is labelled by the mathematics it states.

Finite series. The first prime gap is one; The second prime gap is two; Finite dyadic sums; Finite dyadic sums of consecutive differences; Summation by parts for a rational sequence; The two prime-series normalisations differ by a factor of two; Summation by parts for the primes; Terms of the prime series; Terms of the prime-gap series; The identity relating prime and gap terms.
Convergent series. Convergence of the gap series from convergence of the prime series; The infinite prime-to-gap identity under summability; Prime terms with denominator powers starting at zero; The factor-of-two identity for individual terms; Equivalent irrationality of the prime and gap series; The infinite identity in the second normalisation; Equivalent irrationality in the second normalisation; An elementary polynomial upper bound for the primes; Convergence of the prime series; The prime-to-gap identity with convergence proved.
Recurrences. The recurrence satisfied by rescaled dyadic tails; The difference between two terms of a recurrence; The weighted integer sum accumulated by iteration; Iteration of the recurrence; A shift as a multiple of a tail minus an integer; The recurrence for consecutive tail differences; Subtracting an integer preserves integrality; The criterion for a shift to be integral.
Denominators and integral shifts. The denominator divisibility criterion for an integral shift; Euler's congruence for an odd denominator; An integral shift of totient length; An integral shift remains integral at the next index; An integral shift remains integral at every later index; The denominator after one recurrence step; An odd denominator is unchanged by a step; An even denominator is halved by a step; Removing the power of two from a rational denominator; An integral multiple after the power of two is removed; Eventual integral shifts from the denominator factorisation; Eventual integral shifts under rationality.
Rationality criteria. Rational representation of a rescaled prime-gap tail; The recurrence for that rational representation; Existence of the rational representation under rationality; Rationality is equivalent to one integral positive-length shift; Rationality is equivalent to eventual integrality of a fixed shift; Irrationality is equivalent to nonintegrality of every positive-length shift; Irrationality is equivalent to arbitrarily late nonintegral shifts of each length.
Small differences and counterexamples. Two small differences with unequal coefficients cannot both be integral; Arbitrarily late small unequal pairs exclude eventual integrality; The small-pair implication for prime gaps; Prime gaps exceed every prescribed bound; Prime gaps are not eventually periodic; Coefficients obtained by telescoping; The finite telescoping identity; The telescoping example with coefficients equal to the index minus one; Nonperiodicity of the telescoping example; Rationality forces some fixed shift to fail eventual smallness.
Further finite identities. Finite prime sums in the second normalisation; Adding one term to a dyadic partial sum; Adding one term to a difference partial sum; Consecutive primes are in increasing order; The natural-number gap agrees with subtraction in the rationals; Finite sums of weighted prime gaps; Finite sums for the telescoping coefficients; A nonnegative version of the telescoping coefficients; Agreement of the nonnegative and integer formulas.

References

[1] Paul Erdős, *Sur certaines séries à valeur irrationnelle*. L'Enseignement Mathématique **4** (1958), 93–100, doi:10.5169/seals-34629.

[2] Paul Erdős and Ronald L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*. Monographies de L'Enseignement Mathématique, L'Enseignement Mathématique, 1980.

[3] Paul Erdős and Carl Pomerance, *On the largest prime factors of n and $n+1$*. Aequationes Mathematicae **17** (1978), 311–321, doi:10.1007/BF01818569.

[4] Paul Erdős, *On the irrationality of certain series: problems and results*. New Advances in Transcendence Theory, Cambridge University Press, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[5] Paul Erdős, *Beweis eines Satzes von Tschebyschef*. Acta Litterarum ac Scientiarum Szeged **5** (1932), 194–198.

[6] Paul Erdős and Ernst G. Straus, *On the irrationality of certain Ahmes series*. Journal of the Indian Mathematical Society (N.S.) **27** (1964), 129–133. MR0175848.

[7] Paul Erdős and Ernst G. Straus, *On the irrationality of certain series*. Pacific Journal of Mathematics **55** (1974), no. 1, 85–92, doi:10.2140/pjm.1974.55.85.

[8] Kevin Ford, Ben Green, Sergei Konyagin, James Maynard and Terence Tao, *Long gaps between primes*. Journal of the American Mathematical Society **31** (2018), 65–105, doi:10.1090/jams/876; arXiv:1412.5029v3.

[9] John A. Fridy, *Generalized bases for the real numbers*. The Fibonacci Quarterly **4** (1966), no. 3, 193–201.

[10] A. Ya. Khinchin, *Continued Fractions*. University of Chicago Press, 1964.

[11] Vjekoslav Kovač and Terence Tao, *On several irrationality problems for Ahmes series*. Acta Mathematica Hungarica **175** (2025), 572–608, doi:10.1007/s10474-025-01528-0; arXiv:2406.17593v4. Statement and section numbers refer to arXiv version 4.

[12] Vivian Kuperberg, *Sums of singular series with large sets and the tail of the distribution of primes*. The Quarterly Journal of Mathematics **74** (2023), no. 4, 1457–1479, doi:10.1093/qmath/haad030; arXiv:2210.09775v2. Statement numbers refer to arXiv version 2, 15 June 2023.

[13] Johan Land, *A conditional proof of the irrationality of $\sum_{n\geq 1} p_n 2^{-n}$ under a uniform Hardy–Littlewood prime-tuples conjecture*. Research draft, 5 September 2026. Formalisation material accompanies the draft.

[14] Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*. Automated Deduction – CADE 28, Lecture Notes in Computer Science, Springer, 2021, pp. 625–635, doi:10.1007/978-3-030-79876-5_37.

[15] The mathlib Community, *The Lean mathematical library*. Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs, ACM, 2020, pp. 367–381, doi:10.1145/3372885.3373824. Describes a Lean 3-era snapshot; the repository lock identifies the library used by the present Lean 4 sources.

[16] James Maynard, *Small gaps between primes*. Annals of Mathematics **181** (2015), 383–413, doi:10.4007/annals.2015.181.1.7.

[17] Hugh L. Montgomery and Robert C. Vaughan, *Multiplicative Number Theory I: Classical Theory*. Cambridge Studies in Advanced Mathematics, Cambridge University Press, 2007, doi:10.1017/CBO9780511618314.

[18] Jan-Christoph Schlage-Puchta, *The irrationality of some number theoretical series*. Acta Arithmetica **126** (2007), no. 4, 295–303, doi:10.4064/aa126-4-1; arXiv:1105.1451v1. Published in 2007; arXiv upload is from 2011. Lemma and preprint page locators refer to arXiv version 1.

[19] Ian Short, *Ford Circles, Continued Fractions, and Rational Approximation*. The American Mathematical Monthly **118** (2011), no. 2, 130–135, doi:10.4169/amer.math.monthly.118.02.130; arXiv:0912.1997v1. Preprint title: Ford circles, continued fractions, and best approximation of the second kind. Preprint theorem locators refer to arXiv version 1.

[20] Wouter van Doorn and Vjekoslav Kovač, *Lacunary sequences whose reciprocal sums represent all rational numbers in an interval*. Acta Arithmetica **223** (2026), 275–295, doi:10.4064/aa251001-13-1; arXiv:2509.24971v3. Statement numbers refer to arXiv version 3.

[21] Yitang Zhang, *Bounded gaps between primes*. Annals of Mathematics **179** (2014), 1121–1174, doi:10.4007/annals.2014.179.3.7.

[22] Thomas F. Bloom, *Erdős Problem #251*. 2026. Accessed 6 September 2026.

[23] Erdős Problems contributors, *Erdős Problem #251 discussion thread*. 2026. Accessed 6 September 2026: Tao comment of 7 October 2025 and Land comments of 6 September 2026.

[24] ChatGPT 5.4 Pro (orchestrated by Vjeko Kovač), *On the Erdős problem #251*. Unpublished note, Department of Mathematics, University of Zagreb, 2026. Accessed 6 September 2026.

[25] The Formal Conjectures Authors, *FormalConjectures.ErdosProblems.251*. 2025. Pinned statement source, not a proof of Problem 251; accessed 28 July 2026.

[26] Stefan Ringer, *Local gap statistics, telescoping, and normality: a local-pattern approach to Erdős problem 251*. Preprint, 11 September 2026. Mutable main-branch TeX consulted 16 September 2026; the cited conditional statements were rechecked 18 September 2026. Formalisation material accompanies the draft.

[27] Tonći Crmarić and Vjekoslav Kovač, *On the irrationality of certain super-polynomially decaying series*. Colloquium Mathematicum **179** (2025), 55–68, doi:10.4064/cm9628-5-2025; arXiv:2504.18712v1. Lemma 4 is cited using arXiv version 1.

[28] Boris Adamczewski, Michael Drmota and Clemens Müllner, *(Logarithmic) densities for automatic sequences along primes and squares*. Transactions of the American Mathematical Society **375** (2022), no. 1, 455–499, doi:10.1090/tran/8476; arXiv:2009.14773v2. Theorems 1.2 and 1.4 refer to arXiv version 2, 13 April 2021; journal publication is 2022.

[29] Wouter van Doorn, *Partitions with prescribed sum of reciprocals: asymptotic bounds*. 2025; arXiv:2502.02200v2. Version 2, 23 July 2025.

[30] Szymon Głąb and Franciszek Prus-Wiśniowski, *Achievement sets – current results and open problems*. Real Analysis Exchange (2026), doi:10.14321/realanalexch.1766383782; arXiv:2512.17285v1. Advance publication, first available in Project Euclid 8 June 2026. Consulted text remains arXiv:2512.17285v1 (19 December 2025).

[31] Franciszek Prus-Wiśniowski and Jolanta Ptak, *Achievable Cantorvals almost without reversed Kakeya conditions*. 2024; arXiv:2412.08768v1. Version 1 submitted 11 December 2024. The sparse indices satisfy the overlap inequality, not its strict term-dominating reverse.

[32] Artur Bartoszewicz, Małgorzata Filipczak and Emilia Szymonik, *Multigeometric sequences and Cantorvals*. Central European Journal of Mathematics **12** (2014), no. 7, 1000–1007, doi:10.2478/s11533-013-0396-4; arXiv:1304.4218v2.

[33] Vivian Kuperberg, *Sums of singular series along arithmetic progressions and with smooth weights*. International Journal of Number Theory **21** (2025), no. 1, 53–74,

doi:10.1142/S1793042125500046; arXiv:2301.06095v1. Preprint uploaded 15 January 2023; journal publication is 2025.

[34] Abhishek Jha, *The Poisson Tail Conjecture for primes in short intervals*. 2026; arXiv:2605.23014v2. Substantially revised version 2, 12 September 2026, 31 pages. Conditional statements must retain their strong Hardy–Littlewood hypotheses.

[35] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*. Research in the Mathematical Sciences **1** (2014), article 12, doi:10.1186/s40687-014-0012-7; arXiv:1407.4897v4. Theorem 1.4(i) refers to arXiv version 4 (22 December 2014); the journal numbers it Theorem 4(i). An erratum is recorded at doi:10.1186/s40687-015-0033-x.

[36] Zbigniew Nitecki, *Cantorvals and Subsum Sets of Null Sequences*. The American Mathematical Monthly **122** (2015), no. 9, 862–870, doi:10.4169/amer.math.monthly.122.9.862; arXiv:1106.3779v2. Consulted preprint: Subsum Sets: Intervals, Cantor Sets, and Cantorvals, version 2 (8 July 2013). Theorem 14 is attributed there to Guthrie–Nymann; its locator is not journal pagination.

[37] Piotr Miska, Franciszek Prus-Wiśniowski and Jolanta Ptak, *More on Kakeya Conditions for Achievement Sets*. Results in Mathematics **78** (2023), article 113, doi:10.1007/s00025-023-01890-x. Repairs an estimate in the 2021 proof, preserving its uniqueness conclusion, and gives a simpler proof of a weaker theorem without that conclusion.

[38] Jacek Marchwicki and Piotr Miska, *On Kakeya Conditions for Achievement Sets*. Results in Mathematics **76** (2021), article 181, doi:10.1007/s00025-021-01479-2. Theorem 2.1 is to be read with the proof repair in Miska–Prus-Wiśniowski–Ptak (2023).

[39] Piotr Nowakowski, *On a new condition implying that an achievement set is a Cantorval and its applications*. 2025; arXiv:2512.17761v1. Version 1, 19 December 2025. Theorem 3.1 requires the Star Procedure of Definition 2 never to break; no application to the present factorial weights is asserted.

Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 2.1	Lean · Comparator	Corollary 2.2	Lean · Comparator
Proposition 4.1	Lean · Comparator	Theorem 4.2	Lean · Comparator
Theorem 4.3	Lean · Comparator	Corollary 4.4	Lean · Comparator
Theorem 5.2	Lean · Comparator	Theorem 5.3	Lean · Comparator
Lemma 5.4	Lean · Comparator	Theorem 5.5	Lean · Comparator
Theorem 5.6	Lean · Comparator	Theorem 6.1	Lean · Comparator
Corollary 6.2	Lean · Comparator	Proposition 6.3	Lean · Comparator
Theorem 6.4	Lean · Comparator	Proposition 6.5	Lean · Comparator
Proposition 6.6	Lean · Comparator	Proposition 6.7	Lean · Comparator
Theorem 7.1	Lean · Comparator	Theorem 7.2	Lean · Comparator pending
Theorem 8.1	Lean · Comparator	Proposition 8.2	Lean · Comparator
Theorem 8.3	Lean · Comparator	Corollary 8.4	Lean[†]

Statement	Formal proof	Statement	Formal proof
Proposition 8.5	Lean · Comparator	Corollary 8.6	Lean[†]
Theorem 8.7	Lean[†]	Theorem 8.8	Lean · Comparator
Proposition 8.9	Lean · Comparator	Proposition 8.10	Lean · Comparator
Proposition D.1	Lean · Comparator	Proposition D.2	Lean · Comparator
Proposition D.3	Lean · Comparator	Proposition D.4	Lean · Comparator
Proposition D.6	Lean · Comparator	Theorem E.1	Lean · Comparator